



**Verified Carbon
Standard**

SHANGLUO SHANGZHOU DISTRICT MSW DISPOSAL SITE BIOGAS POLLUTION CONTROL AND COMPREHENSIVE UTILIZATION PROJECT



Project title	Shangluo Shangzhou District MSW Disposal Site Biogas Pollution Control and Comprehensive Utilization Project
Project ID	5314
Crediting period	23-Mar-2023 to 22-Mar-2033 (both days included)
Original date of issue	21-Oct-2024
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1 PROJECT DETAILS

1.1 Summary Description of the Project

Shangluo Shangzhou District MSW Disposal Site Biogas Pollution Control and Comprehensive Utilization Project (hereafter referred to as the project) is located at Shangluo MSW treatment site, Longshan Village, Dazhaoyu Sub-district Office, Shangluo City, Shaanxi Province, P.R. China. The project is constructed and operated by Shangluo BCCY New Power CO., Ltd. (hereafter referred to as the project owner).

Shangluo MSW treatment site is located at Longshan Village, Dazhaoyu Sub-district Office, Shangluo City, Shaanxi Province, P.R. China, and started operation in 2008 with a total design effective capacity of 1,900,000m³.

The project started to construct on 21-Dec-2022. The project has a total designed capacity 1MW (2*0.5MW) and combusts the LFG, which contains nearly 50% of methane, to generate electricity and export it to the Northwest China Power Grid (NCPG). Total electricity supplied to the grid in the crediting period from 23-Mar-2023 to 22-Mar-2033 will be 52,696MWh. VCU's will be claimed from the methane destroyed and the grid electricity displaced. The estimated annual average GHG emission reductions of the project is 32,121tCO₂e and the total GHG emission reductions and removals are 321,210tCO₂e during the fixed 10 years crediting period from 23-Mar-2023 to 22-Mar-2033.

Scenario existing prior to the implementation of the project (the same as baseline scenario):

LFG from Shangluo MSW treatment site is emitted to the atmosphere directly.

Equivalent electricity generated by the project is supplied by the Northwest China Power Grid (NCPG), which is dominated by fossil fuel based power plants.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	This PD(version 01) is used for publishment, the Validation date is not yet determined.	Verified Carbon Standard	LGAI Technological Center,S.A.(Applus+ Certification)	/

1.3 Sectoral Scope and Project Type

Sectoral scope¹	1;13
Project activity type	Energy industries (renewable/non-renewable) ; Waste handling and disposal

This section is not applicable as the project is not a AFOLU project.

Sectoral scope	NA
AFOLU project category²	NA
Project activity type	NA

1.4 Project Eligibility

1.4.1 General eligibility

As per section 2.1.1 of VCS Standard (version 4.7), the scope of the VCS Program includes:

VCS standard scope	Fulfilled (Yes/No)	Justification
1) The seven Kyoto Protocol greenhouse gases.	Yes	The project is a power generation project by combusting the captured landfill gas, which contains nearly 50% of methane. Thus, the project is applicable to this scope.
2) Ozone-depleting substances (ODS).	No	The project activity does not involve any Ozone-depleting substances.
3) Project activities supported by a methodology approved under the VCS Program through the methodology development and review process.	Yes	The project activity is supported by the CDM methodology.
4) Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions).	Yes	The applied methodologies AMS-III.G-Landfill methane recovery (Version 10.0) and AMS-I.D-Grid connected renewable electricity generation (Version 18.0) are methodologies approved under CDM Program, which is also a VCS approved GHG program.

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

² See Appendix 1 of the VCS Standard

5)Jurisdictional REDD+ programs and nested REDD+ projects as set out in the Jurisdictional and Nested REDD+ (JNR) Requirements.	No	This is not a Jurisdictional REDD+ program nor a nested REDD+ project, hence this eligibility criterion is not applicable.
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Furthermore, the project does not belong to the projects excluded in Table 1 of VCS Standard (Version 4.7).

According to 3.1 of VCS standard (Version 4.7), establishing a consistent and standardized certification process is critical to ensuring the integrity of VCS projects. Accordingly, certain high-level requirements must be met by all projects, as set out below.

Requirements of 3.1 of VCS standard (version 4.7)	Justification
3.1.1 Projects shall meet all applicable rules and requirements set out under the VCS Program, including this document. Projects shall be guided by the principles set out in Section 2.2.1.	The project meets all applicable rules and requirements set out under the VCS Program, including VCS standard (version 4.7).
3.1.2 Projects shall apply methodologies eligible under the VCS Program. Methodologies shall be applied in full, including the full application of any tools or modules referred to by a methodology, noting the exception set out in Section 3.14.1. The list of methodologies and their validity periods is available on the Verra website.	The project applies the methodologies AMS-III.G-Landfill methane recovery (Version 10.0) and AMS-I.D-Grid connected renewable electricity generation (Version 18.0) along with tools or modules as applicable developed under the United Nations Clean Development Mechanism, which can be used for projects and programs registering with VCS as per requirements available on the Verra website.
3.1.3 Projects shall apply the latest version of the applicable methodology in all cases unless a grace period applies to the project as set out in 3.22 below. Projects shall update to the latest version of the methodology when reassessing the baseline or renewing a crediting period.	The project applies the methodologies AMS-III.G-Landfill methane recovery (Version 10.0) and AMS-I.D-Grid connected renewable electricity generation (Version 18.0), which are the latest version.
3.1.4 Projects and the implementation of project activities shall not lead to the violation of any applicable law, regardless of whether or not the law is enforced.	According to the environment impact assessment report (EIA) of the project, the project complies with all Chinese relevant laws and regulations.

3.1.5 Where projects apply methodologies that permit the project proponent its own choice of model (see the VCS Program Definitions for the definition of model), the model shall meet the requirements set out in the VCS Methodology Requirements, and it shall be demonstrated at validation that the model is appropriate to the project circumstances (i.e., use of the model will lead to an appropriate quantification of GHG emission reductions or carbon dioxide removals).	Not applicable as there is no model needs to be chosen by the project proponent as per the applied methodologies AMS-III.G-Landfill methane recovery (Version 10.0) and AMS-I.D-Grid connected renewable electricity generation (Version 18.0).
3.1.6 Where projects apply methodologies that permit the project proponent to choose a third-party default factor or standard to ascertain GHG emission data and any supporting data for establishing baseline scenarios and demonstrating additionality, such default factor or standard shall meet the requirements set out in the VCS Methodology Requirements.	Not applicable as there are no third-party default factor or standard need to be chosen by the project proponent as per the applied methodologies AMS-III.G-Landfill methane recovery (Version 10.0) and AMS-I.D-Grid connected renewable electricity generation (Version 18.0).
3.1.7 Where the rules and requirements under an approved GHG program conflict with the rules and requirements of the VCS Program, the rules and requirements of the VCS Program shall take precedence.	Not applicable. The project obeys the rules and requirements of the VCS Program
3.1.8 Where projects apply methodologies from approved GHG programs, they shall conform with any specified capacity limits (see the VCS Program Definitions for the definition of capacity limit) and any other relevant requirements set out with respect to the application of the methodology and/or tools referenced by the methodology under those programs.	Not applicable as there is no specified capacity limits or any other relevant requirements set out as per the applied methodology and tools.
3.1.9 Where Verra issues new VCS Program rules, the effective dates of these requirements are set out in Appendix 3 Document History and Effective Dates or equivalent for other program documents, and are listed in a companion	The project has followed the latest requirements of Verra.

Summary of Effective Dates document which corresponds with each update.	
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1.4.2 AFOLU project eligibility

The project is not a AFOLU project. Thus, this section is not applicable.

1.4.3 Transfer project eligibility

The project is not a transfer project. Thus, this section is not applicable.

1.5 Project Design

- ☒ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☐ Grouped project

1.5.1 Grouped project design

The project is not a grouped project. Thus, this section is not applicable.

1.6 Project Proponent

Organization name	Henan BCCY Environmental Energy Co., Ltd.
Contact person	Lei Wang
Title	Minister of carbon emission department
Address	10/F, Kailin Building, 99 Ruyi West Road, No.22, Zhengzhou, Henan Province, P.R. China
Telephone	+86 371 65521780
Email	lwang@bccynewpower.com

1.7 Other Entities Involved in the Project

Organization name	Shangluo BCCY New Power CO., Ltd.
Role in the project	Project Operation Entity
Contact person	Yongmei Fang

Title	Project manager
Address	10/F, Kailin Building, 99 Ruyi West Road, No.22, Zhengzhou, Henan Province, P.R. China
Telephone	+86 371 56737901
Email	ymfang@bccynewpower.com

1.8 Ownership

The project owner is Shangluo BCCY New power CO., Ltd. The approval of Feasibility Study Report (FSR) and Environmental impact Assessment (EIA), and the business license of the project owner are the evidences for legislative right. Besides, the equipment purchasing contract and the purchasing power agreement are the evidences for the ownership of the plant equipment and power generating. Shangluo BCCY New power CO., Ltd. is a wholly-owned subsidiary of Henan BCCY Environmental Energy Co., Ltd., which is responsible for the acquisition of approvals, construction and operation of project. Henan BCCY Environmental Energy Co., Ltd. is authorized as the project Proponent by Shangluo BCCY New power CO., Ltd.

1.9 Project Start Date

Project start date	23-Mar-2023
Justification	The project started to operate on 23-Mar-2023, which is line with the requirement of the VCS standard (version 4.7) - The project start date of a NON-AFOLU project is the date on which activities that lead to the generation of reductions or removals are implemented.

1.10 Project Crediting Period

Crediting period	☐ Seven years, twice renewable ☒ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	23-Mar-2023 to 22-Mar-2033

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☒ < 300,000 tCO₂e/year (project)

☐ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
23-Mar-2023 to 31-Dec-2023	19,216
01-Jan-2024 to 31-Dec-2024	26,176
01-Jan-2025 to 31-Dec-2025	27,672
01-Jan-2026 to 31-Dec-2026	29,208
01-Jan-2027 to 31-Dec-2027	30,791
01-Jan-2028 to 31-Dec-2028	32,431
01-Jan-2029 to 31-Dec -2029	34,138
01-Jan-2030 to 31-Dec-2030	35,906
01-Jan-2031 to 31-Dec-2031	37,745
01-Jan-2032 to 31-Dec-2032	39,650
01-Jan-2033 to 22-Mar-2033	8,277
Total estimated ERRs during the first or fixed crediting period	321,210
Total number of years	10
Average annual ERRs	32,121

1.12 Description of the Project Activity

GHG emission reductions will be achieved through combustion of the recovered methane gas by gas engines, which would be otherwise emitted to the atmosphere, and the generation of electricity from the landfill gas, which is supplied by the Northwest China Power Grid (NCPG) prior to the implementation of the project.

Description of the technology in the project

The project consists in LFG collection, pre-treatment system, with subsequent electricity generation and grid connection system.

LFG collection system

LFG is extracted from MSW under negative pressure by blower pumps and moved through wells, then collected by gas collection stations and transferred by sub-pipes and a main pipe to LFG treatment equipment. Flow rate of the LFG is regulated at the collection points in order to always fit with the consumption capacity of the generation engines.

LFG pre-treatment system

Prior to electricity generation, LFG is pre-treated to remove impurities and moisture, to avoid corrosion in the engines. The treatment consists of filtration, de-moisturing, cooling and pressurization.

Electricity generation and grid connection system

2 gas engines of 0.5MW rated power each are fed with the LFG and generate electricity, which is then exported to the grid.

Table 1.1 Equipment technical parameters³

Main Equipment	Parameter	Value
1# and 2# generator set	Type	500GJ1-PWZ-TEM
	Manufacturer	SHENGLI OIL FIELD SHENGLI POWER MACHINERY GROUP CO.,LTD
	Rated capacity	500 kW
	Rated rotation speed	1000 r/min
	Rated frequency	50 Hz

1.13 Project Location

The project is located at Shangluo MSW treatment site, Longshan Village, Dazhaoyu Sub-district Office, Shangluo City, Shaanxi Province, P.R. China. The central geographical coordinates of the project site are longitude 110°09'51.600"E and latitude 34°04'22.270"N. Figure 1.1 shows the location of the Project.

³ Sourced from generator set nameplate.

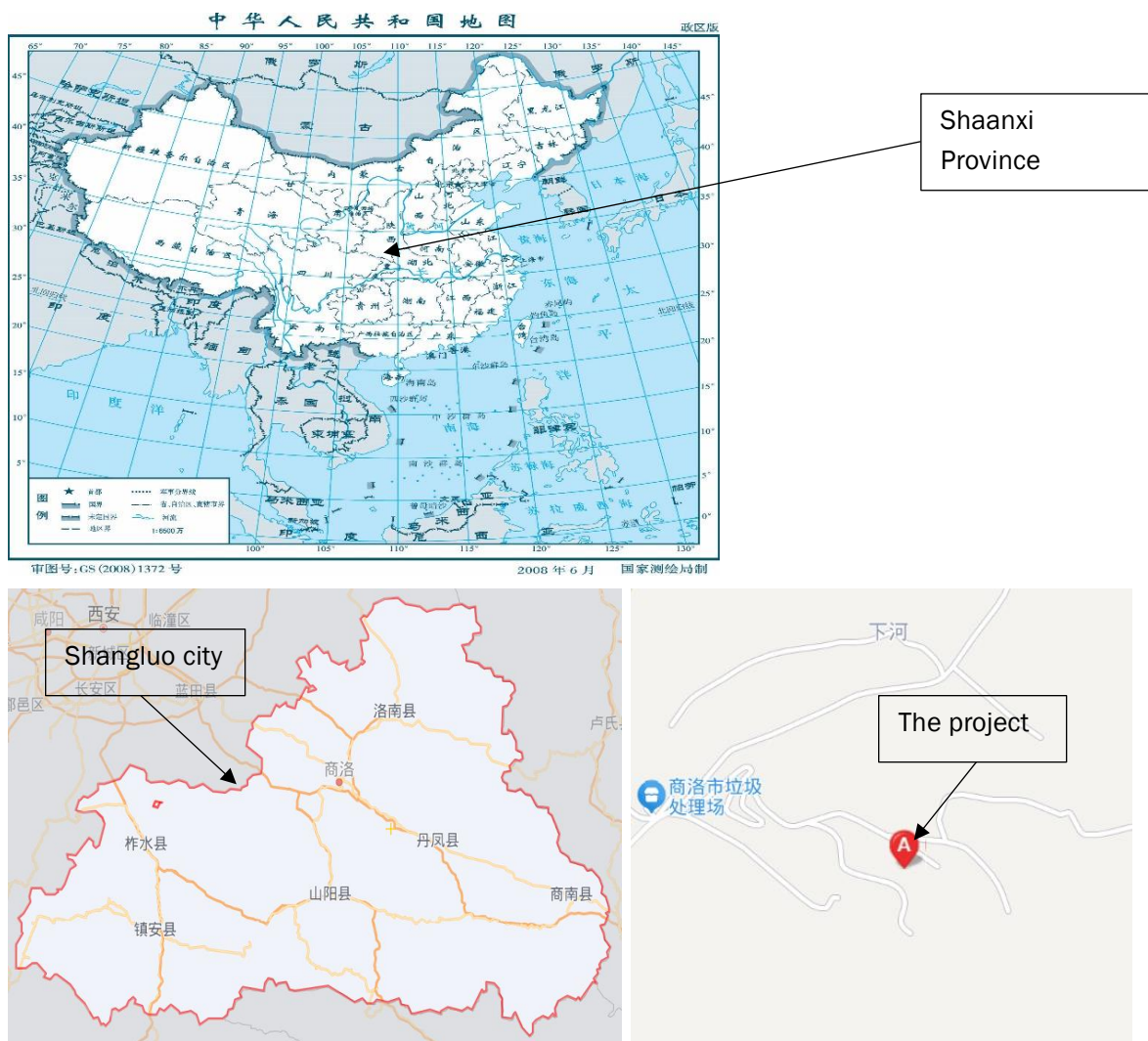


Figure 1.1 Location of the project

1.14 Conditions Prior to Project Initiation

The baseline scenarios are the same as the conditions existing prior to the project initiation. The details are showed in the Section 3.4.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project is in the field of waste handling and disposal and meets the relevant provision of the law of the People's Republic of China on the Prevention and Control of Environmental Pollution by Solid Waste. The generation of power exported to NCPG will meet the requirement of national laws and regulations, also financially viable.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

1.18 Sustainable Development Contributions

Shangluo Shangzhou District MSW Disposal Site Biogas Pollution Control and Comprehensive Utilization Project is located at Shangluo MSW treatment site, Longshan Village, Dazhaoyu Sub-district Office, Shangluo City, Shaanxi Province, P.R. China, whose objective is to collect the landfill gas (LFG) from Shangluo MSW treatment site and utilize it to generate electricity. The project involves the installation of LFG collection, pre-treatment system, with subsequent electricity generation and grid connection system.

The project will contribute to Sustainable Development Goals in the following aspects:

SDG 7 Affordable and Clean Energy

The project utilizes the LFG from the Shangluo MSW treatment site to generate electricity by gas engines. Total electricity generation supplied by the project to the grid is estimated to be 52,696MWh, which could replace the equivalent electricity from fossil fuel based grid.

SDG 8 Decent works and Economic Growth

The project can provide full time job opportunities over the project lifetime.

SDG 13 Climate Action

The project utilizes the LFG for electricity generation, which contains nearly 50% of methane, to substitute the equivalent electricity supplied by the grid. Total GHG emission reductions and removals of 321,210tCO₂e will be achieved during the fixed 10 years crediting period.

Sustainable development priorities are illustrated in the People's Republic of China National Report on Sustainable Development issued on 01-June-2012. Sustainable development priorities of the Report related to the project are promoting new energy development, strengthening environmental pollution control and responding to climate change. Methane is an ideal clean fuel. Each cubic meter of methane contains about 360000kJ calorific value, LFG recovery and utilization will supplement the energy supply of the grid and contributes to the new energy development in China. Odour from landfill negatively impacts on residents around the landfill. Implementation of the project will reduce odours through LFG collection and will thus mitigate the impact of landfill odour on people's daily lives. Landfill gas contains nearly 50% of methane and methane is a greenhouse gas some 28 times more potent than carbon dioxide. LFG recovery and utilization is contribution to mitigating global warming.

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

There is no equipment transferred in the project, no leakage effects need to be accounted under AMS-III.G and AMS-I.D.

1.19.2 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the Project Description

1.19.3 Further Information

Not applicable.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	The purpose of the project is to utilize the captured LFG for power generation, which is then delivered the electricity to the Northwest China Power Grid (NCPG). The entire project must adhere to local regulations, rules, and obtain necessary permissions. Throughout detailedly analyzing the geographic boundary, technology process, potential pollution, potential environmental impacts to identify individuals and organizations with a vested interest or potential influence over the project. The following people are considered as the stakeholders of this project during the implementation stage. Residents of the nearby village. Employees of Shangluo MSW treatment site. Relevant administrative staff of the local government. Employees of Shangluo BCCY New Power CO., Ltd. The Project Proponent.
Legal or customary tenure/access rights	According to the cooperation agreement signed between the project owner and Shangluo Shangzhou District MSW Disposal Site, Shangluo Shangzhou District MSW Disposal Site will provide

	production and living land for project owner. So the project does not involve the land issues.
Stakeholder diversity and changes over time	Stakeholders identified for this project activity are diverse in nature – in terms of socio-economic status, age and gender wise and the diversity is unlikely to change over time.
Expected changes in well-being	The project is expected to improve the economic and environmental well-being as below. 1. Economic well-being – the project activity is likely to provide employments to the locality thus economic well beings are expected; 2. Environmental well-being – (1) as this project activity reduces the methane emissions in atmosphere by recovering and utilizing LFG to generate electricity, it will bring environmental well-being in the locality. (2) Total electricity generation supplied by the project to the grid is estimated to be 52,696MWh, which could replace the equivalent electricity from fossil fuel based grid.
Location of stakeholders	The project locates at Longshan Village, Dazhaoyu Sub-district Office, Shangluo City, Shaanxi Province, P.R. China. And the stakeholders involved in the project located at Longshan Village, Dazhaoyu Sub-district Office, Shangluo City, Shaanxi Province, P.R. China. Since the project does not involve activities outside the project boundaries, areas outside the project area will not be affected by the project.
Location of resources	The project is not involved the territories and resources, or customary access of the stakeholders own.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	30-Nov-2022
Stakeholder engagement process	The stakeholders were informed about the stakeholder meeting through posters on 15-Nov-2022. In the bulletin, the company invited the potential stakeholders to get to know their opinions and/or suggestions about the implementation of the project and

	VCS application of the project. The stakeholder consultancy meeting for the parties interested in the project was organized at Shangluo MSW treatment site on 30-Nov-2022 to collect opinions from all stakeholders, such as representatives of local residents, employees of Shangluo MSW treatment site, relevant administrative staff of the local government, employees of Shangluo BCCY New Power CO., Ltd. and Project Proponent. The project owner appointed one person to make a meeting minute. Bulletin and meeting are in Chinese. The stakeholder meeting was attended by 15 people, four of whom were women.
Consultation outcome	During the stakeholder consultancy meeting, some stakeholders worried that the project will bring noise, land occupying and pollution on employees and local residents' living. For these issues, the project owner will offer some measures: i.e. installation of sound proof devices and plating of green isolation belts are used for mitigating noise; the area the project built on only will not occupy farms of local residents; at the project site, waste recycling will be carried out reduce emit of waste, and the project site meanwhile will be far from the residential area, so the effect of the noise from the plant is little to local residents. All attendances to the meeting were satisfied to the measures. In general, all participants have positive perception on project design and implementation. The project would actually facilitate the development of the local economy and increase the income of local residents. During the stakeholder consultancy meeting, the project owner assured everyone that they would comply with labor laws and ensure the legitimate rights and interests of workers. The project owner also introduced the VCS validation and verification process to everyone.
Ongoing communication	During the stakeholder consultancy meeting, grievance mechanism was also explained. The project owner will regularly hold stakeholder meetings, distribute questionnaires and give the response to various stakeholders. A grievance expression book was put in the office of the project, which is used to collect stakeholders' views about the project. Project Participant checked the comments in the book on a regular basis, and record responses. The project information with contact information was also posted on the bulletins at and nearby the project site. Every stakeholder can make comments by grievance expression book or contact information. Project Participant will be respectful to the views of stakeholders

	and suggest alternative solutions or compromises wherever possible. Until now, no comments or Input / Grievance Expression received from stakeholder.
Stakeholder input	The project did not receive major negative input during the consultation. The project design need not to have any updates. The project would actually facilitate the development of the local economy and increase the income of local residents. All stakeholders were pleased with the development of the project.

2.1.3 Free Prior and Informed Consent

Obtaining consent	Based on the principle of friendly consultation and on the basis of truly and fully expressing their respective wishes, the project owner and Shangluo Shangzhou District MSW Disposal Site signed a cooperation agreement on 13-Jul-2022 regarding to this project. Based on questionnaire surveys and stakeholder consultation meetings, local residents have expressed support for the project. The feasibility study report of the project was approved by the Shangzhou District Administrative Approval Service Bureau of Shangluo on 25-Jul-2022. An environmental impact assessment (EIA) of the project was approved by Shangluo Branch of Shangluo Ecological Environment Bureau on 14-Dec-2022. There are not ongoing or unresolved conflicts.
Outcome of FPIC	The feasibility study report of the project was approved by the Shangzhou District Administrative Approval Service Bureau of Shangluo on 25-Jul-2022. An environmental impact assessment (EIA) of the project was approved by Shangluo Branch of Shangluo Ecological Environment Bureau on 14-Dec-2022. There are not ongoing or unresolved conflicts. The project is located at Shangluo MSW treatment site, Longshan Village, Dazhaoyu Sub-district Office, Shangluo City, Shaanxi Province, P.R. China. On 13-Jul-2022, the project owner and Shangluo Shangzhou District MSW Disposal Site signed a cooperation agreement on this project. According to the project

	cooperation agreement, the project has the right to use the land, and there is no violation of the rights of other third parties. Based on questionnaire surveys and stakeholder consultation meetings, local residents have expressed support for the project. The project was constructed with the consent of government departments at all levels. The project has not encroached on land, relocated people without consent, and forced physical or economic displacement.
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2.1.4 Grievance Redress Procedure

Development process	A grievance expression book was put in the office of the project, which is used to collect stakeholders' views about the project. Project Participant checked the comments in the book on a regular basis, and record responses. The project information with contact information was also posted on the bulletins at and nearby the project site. Every stakeholder can make comments by grievance expression book or contact information. Project Participant will be respectful to the views of stakeholders and suggest alternative solutions or compromises wherever possible.
Grievance redress procedure	Once a comment/complaint is received, project proponent will firstly discuss the possible solutions. Then project proponent will hold a meeting to communicate with the stakeholders as soon as possible. After the meeting, Project proponent will provide a written response stated the feasible solutions. Project proponent will try their best efforts to resolve complaints within a reasonable period of time and make the stakeholders satisfy the solutions.

2.1.5 Public Comments

Comments received	Actions taken
NA	NA

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The project proponent has successfully operated several recovery to power plants in China. It has also successfully developed carbon projects related to LFG recovery to power under VCS

programme (VCS reference number: 2353, 2731, 2840). Therefore, the PP has expertise and experience in implementing similar project activities.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No risk identified	The investment, construction and operation of the project are strictly in compliance with the relevant laws and regulations of the People's Republic of China. The project is a LFG recovery to power project. After implementation of the project, it can provide economic and environmental well-being to stakeholders.
Risks to stakeholder participation	No risk identified	Based on the stakeholder meeting, the stakeholders involved in the project are very supportive of the project. The investment, construction and operation of the project are strictly in compliance with the relevant laws and regulations of the People's Republic of China.
Working conditions	No risk identified	The investment, construction and operation of the project are strictly in compliance with the relevant laws and regulations of the People's Republic of China. The project has DCS system. Most of the employees work in indoor environment (at the office), instead of having to stand outside. In case of on-site monitoring and device maintenance, the project owner provides safety protection equipment.
Safety of women and girls	No risk identified	The investment, construction and operation of the project are strictly in compliance with the relevant laws and regulations of the People's Republic of China. China has published "Women's Rights Protection Law" ⁴ to provide support and protection to women and girls.

⁴ https://www.gov.cn/guoqing/2021-10/29/content_5647634.htm

Safety of minority and marginalized groups, including children	No risk identified	The investment, construction and operation of the project are in compliance with the relevant laws and regulations of the People's Republic of China. China has published protection laws on the minor⁵. Employment regulations are described in this Law. The Project requires skilled employees to operate, maintain, and manage the power plant. Therefore, the project does not employ and is not complicit in any form of child labour. According to the Employee list, it could be confirmed that the Project does not employ any child labour.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	No risk identified	An environmental impact assessment (EIA) of the project was approved by Shangluo Branch of Shangluo Ecological Environment Bureau on 14-Dec-2022. Measures has been identified in EIA of the project to mitigate environment pollution caused by construction and operation of the project, which will be implemented as per environment pollution related laws and regulations and supervised by local government. During construction phase, the raise dust, noise, waste water and solid wastes caused by project construction will be treated according to the measures in EIA report, and there will be no significant impact on the environment. During the operation period, all the mitigation measures proposed by EIA report will be implemented and the following key aspects will be addressed: Waste Gas: The main source of exhaust gas pollution in this project is the exhaust gas from the generator set. Each generator set is equipped with a 15m high exhaust funnel for high-altitude emissions. The exhaust gas can meet the limits listed in

⁵ http://www.gov.cn/xinwen/2020-10/18/content_5552113.htm

Table 2 of “Integrated Emission Standard of Air Pollutants” (GB16297-1996). It will not have a significant impact on the surrounding environment.

Waste water:

The waste water of the project is mainly condensate and domestic wastewater. After pre-treatment in septic tanks, domestic sewage will be transported together with condensed water to the leachate treatment station of Shangzhou District landfill site, and finally enter to the Shangluo City sewage treatment plant for centralized treatment. The project won't have a negative impact on the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s).

Noise pollution:

The noise is mainly generated by the generator set, pump equipment and other equipment. Through the adoption of low noise equipment, foundation shock absorption, building sound insulation, muffler, equipment daily maintenance and other measures, the noise contribution value of the factory boundary can meet the requirements of class 2 of Industrial Enterprise factory boundary Environmental noise Emission Standard (GB12348-2008). The impact of production noise on the surrounding environment is relatively small.

Solid waste:

The solid waste of the project is mainly general solid waste, waste filter residue, waste desulfurizer, waste engine oil and waste catalyst. The general solid waste and waste filter residue is collected and transported to landfill site for landfilling. The waste desulfurizer will be replaced by the manufacturer and uniformly recycled. The

		waste engine oil and waste catalyst are belonging to hazardous waste, which will be stored in the hazardous waste room of the factory, and regularly handed over to qualified units for recycling and processing. The solid waste will not have a significant impact on the surrounding environment.
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2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified ⁶	Mitigation or preventative measure(s) taken
Discrimination	No risk identified	China has published “Labor Law” ⁷ and “Women’s Rights Protection Law”, which could prohibit such phenomena. The project strictly abides by the rules of equality and does not involve and is not complicit in any form of discrimination.
Sexual harassment	No risk identified	China has published “Labor Law” and “Women’s Rights Protection Law”, which could prohibit such phenomena. The project strictly abides by the rules of equality and does not involve and is not complicit in any form of sexual harassment.
Equal pay for equal work	No risk identified	The project proponent strictly abides by the labor law, and sign contracts with employees. Salaries are delimited according to different positions. The legitimate rights and interests of employees are protected by Chinese law. In case of disputes, employees may apply to the local government for labor arbitration.
Gender equity in labor and work	No risk identified	The project has male and female employees of different genders. There is no significant gender difference between female employees and male employees in terms of salary, promotion opportunities and welfare benefits, and employees of different genders have equal

⁶ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

⁷ https://www.gov.cn/banshi/2005-05/25/content_905.htm

		opportunities to obtain training and development resources.
Forced labor	No risk identified	The project proponent strictly abides by the labor law, established reasonable labor safeguards to protect the working environment and hours of employees and ensure that employees are not forced to work. The project is supervised by local labor authorities.
Child labor	No risk identified	China has published protection laws on the minor. Employment regulations are described in this Law. The Project requires skilled employees to operate, maintain, and manage the power plant. Therefore, the project does not employ and is not complicit in any form of child labor. According to the Employee list, it could be confirmed that the Project does not employ any child labor.
Human trafficking	No risk identified	The project proponent strictly abides by the labor law, the project proponent will verify the identity of the employees, and if unidentified workers are found, the project proponent will report to the local police.

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	Human rights protection is formally embodied in the regulations in China. China fully recognizes international human rights law, and the United Nations Declaration on the Rights of Indigenous Peoples and ILO Convention 169 on Indigenous and Tribal Peoples. China has its own legislation in place prohibiting the violation of human rights principle and it actively enforces the compliance of such principle. China has ratified two UN human rights treaties—the International Covenant on Civil and Political Rights, the International Covenant on Economic, Social, and Cultural Rights. China has published relevant regulations and practice into line with the treaties, such as Labor Law. This project does not discriminate with regards to participation and inclusion. The project abides the rules of equality accordingly and does not involve and is not complicit in any form of discrimination. Specific to

gender equality, detailed enforcement rules are regulated in “Women’s Rights Protection Law” to provide support and benefits to women. The project proponent and its activities are strictly regulated by local and national regulations, which obligate the project activities to in line with the international human right law and conventions. The legal approval by authorities demonstrate that the project does not and will not violate the regulations. According to the relevant provisions of the Labor Law and Women’s Rights Protection Law of China, the project continues to recognize, respect, and promote the protection of the rights of IPs, LCs, and customary rights holders.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No risk identified	According to the EIA report, no indigenous peoples present in or within the area of influence of the project. And the project area does not include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	The project was implemented inside the Shangluo MSW treatment site which was managed by local government. The project has been approved by the local government. And by implementing the project, the local environment has been improved, such as odour from landfill, fire and explosion risk from LFG, etc. The project does not result in any risks related to protecting and preserving the property rights of IPs, LCs, and customary rights holders, and to protecting legal or customary tenure/access rights to territories, property, and resources.

2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	There is no considerable negative effect of project activities to any stakeholder groups. The project was implemented inside the Shangluo MSW treatment site which was managed by local government. The project has been approved by the local government. And by implementing the project, the local environment has been improved, such as odour from landfill, fire and explosion risk from LFG, etc. The project does not result in any risks related to protecting and preserving the property rights of IPs, LCs, and customary rights holders, and to protecting legal or customary tenure/access rights to territories, property, and resources. Therefore, no benefit sharing plan is required.
Summary of the benefit sharing plan	As clarified above, no benefit sharing plan is required for the project.
Approval and dissemination of benefit sharing plan	As clarified above, no benefit sharing plan is required for the project.

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	No risk identified	The project's purpose is to supply electricity from LFG. According to EIA, the project does not involve impacts on biodiversity and ecosystems.
Soil degradation and soil erosion	No risk identified	The EIA report states that a soil impact assessment is not required for this type of project.
Water consumption and stress	No risk identified	Based on the EIA, the total waste water of the project is 316.03m ³ /a. The wastewater of this project will be discharged after treatment to meet the standards, as detailed in 2.2.2.

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes

☒ No

2.4.2 Introduction of Species

N/A. According to EIA, no planting or species introduction are in the project. This section is left as N/A.

Species introduced	Classification	Justification for use	Adverse effects and mitigation
N/A	N/A	N/A	N/A

According to EIA, no invasive species exist in the project area. Therefore, the following table isn't applicable.

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
N/A	N/A

2.4.3 Ecosystem Conversion

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion	No risk identified	Not applicable as the project is not a ARR, ALM, WRC or ACoGS project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AMS-III.G	Landfill methane recovery	10.0
Methodology	AMS-I.D	Grid connected renewable electricity generation	18.0
TOOL	TOOL04	Emissions from solid waste disposal sites	08.1
TOOL	TOOL07	Tool to calculate the emission factor for an electricity system	07.0
TOOL	TOOL32	Positive lists of technologies	04.0

3.2 Applicability of Methodology

The selected methodology AMS-III.G. (version 10.0) and AMS-I.D. (version 18.0) are appropriate to LFG project activities, where the baseline scenario is the atmospheric release of the LFG, and that all or part of the electricity exported to the grid is the electricity generation in existing and/or new grid-connected power plants. In this case, the LFG is released to atmosphere prior to the implementation of the project, electricity generated by the project is exported to Northwest China Power Grid (NCPG).

The project fulfils the following applicability conditions of the methodology:

Methodology ID	Applicability condition	Justification of compliance
AMS-III.G	2.This methodology comprises measures to capture and combust methane from landfills (i.e. solid waste disposal sites) used for the disposal of residues from human activities including municipal, industrial, and other solid wastes containing biodegradable organic matter.	Applicable. The project consists in capturing landfill gas (which contains methane) from the Shangluo MSW treatment site, which is used for disposal of residues from human activities.
	3.Different options to utilise the recovered landfill gas as detailed in paragraph 4 for “AMS-III.H.: Methane recovery in wastewater treatment” (version 19.0) are eligible for use under this methodology. The relevant procedures in AMS-III.H. shall be followed in this regard. Paragraph 4 of AMS-III.H.: The recovery biogas from the above measures may also be utilised for the following applications instead of combustion/flaring: (a) Thermal or mechanical, electrical energy generation directly; (b) Thermal or mechanical, electrical energy generation after bottling of upgraded biogas, in this case additional guidance provided in the appendix shall be followed; or	Applicable as (a).

	(c) Thermal or mechanical, electrical energy generation after upgrading and distribution, in this case additional guidance provided in the appendix shall be followed: (i) Upgrading and injection of biogas into a natural gas distribution grid with no significant transmission constraints; (ii) Upgrading and transportation of biogas via a dedicated piped network to a group of end users; or(iii) Upgrading and transportation of biogas (e.g. by trucks) to distribution points for end users (d) Hydrogen production; (e) Use as fuel in transportation applications after upgrading.	
	4.Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO₂ equivalent annually from all Type III components of the project activity.	Applicable. The project results in aggregate emission reduction of less than 60kt CO₂ equivalent annually from all Type III components.
	5.The proposed project activity does not reduce the amount of organic waste that would have been recycled in the absence of the project activity	The implementation of the project does not reduce the amount of organic waste that would be recycled in the absence of the project. All the solid waste is disposed in the Shangluo MSW treatment site.
	6.This methodology is not applicable if the management of the solid waste disposal site (SWDS) in the project activity is deliberately changed in order to increase methane generation compared to the situation prior to the implementation of the project activity (e.g.	Not applicable.

	other than to meet a technical or regulatory requirement). Such changes may include, for example, the addition of liquids to a SWDS, pre-treating waste to seed it with bacteria for the purpose of increasing the rate of anaerobic degradation of the SWDS or changing the shape of the SWDS to increase methane production	
AMS-I.D	4.This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant (s); (c) Involve a retrofit of (an) existing plant (s); (d) Involve a rehabilitation of (an) existing plant (s)/unit (s) or; Involve a replacement of (an) existing plant(s).	Applicable. The project installs a new power plant at a site where there was no renewable energy power plant operating prior to the implementation of the project, which corresponds to point (a).
	5.Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4W/m²; The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4W/m².	Not applicable. The project is not a hydro power plant.
	6.If the new unit has both renewable and non-renewable components (e.g. a wind/ diesel	Not applicable. The project does not use non-

	unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel. the capacity of the entire unit shall not exceed the limit of 15 MW.	renewable components nor co-fires fossil fuels). Anyway, the total installed capacity is below 15MW.
	7.Combined heat and power (co-generation) systems are not eligible under this category.	Not applicable. The project only involves electricity generation.
	8.In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.	Not applicable. The project does not involve addition of renewable energy generation units at an existing renewable power generation facility.
	9.In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project the total output of the retrofitted, rehabilitated or replacement power plant / unit shall not exceed the limit of 15 MW.	Not applicable.
	10.In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as AMS-L.C. Thermal energy production with or without electricity shall be explored.	Applicable. The project is designed to install two engines to combust LFG of Shangluo MSW treatment site, which mainly contains 50% methane.
	11.In case biomass is sourced from dedicated plantations, the applicability criteria in the tool "Project emissions from cultivation of biomass "shall apply.	Not applicable.

TOOL07	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	Applicable. This tool is applied to estimate the OM, BM and/or CM.
	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	Applicable. The emission factor for the project electricity system is calculated for grid power plants only.
	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Not applicable. The project electricity system is located in China.
	Under this tool, the value applied to the CO2 emission factor of biofuels is zero.	Not applicable.
TOOL04	(a)Application A: The CDM project activity mitigates methane emissions from a specific	Applicable. The project collects the methane from

	existing SWDS. Methane emissions are mitigated by capturing and flaring or combusting the methane (e.g. "ACM0001: Flaring or use of landfill gas). The methane is generated from waste disposed in the past, including prior to the start of the CDM project activity. In these cases, the tool is only applied for an ex ante estimation of emissions in the project design document (CDM-PDD). The emissions will then be monitored during the crediting period using the applicable approaches in the relevant methodologies (e.g. measuring the amount of methane captured from the SWDS)	the Shangluo MSW treatment site and use it to generate the electricity.
	(b)Application B: The CDM project activity avoids or involves the disposal of waste at a SWDS. An example of this application of the tool is ACM0022, in which municipal solid waste (MSW)is treated with an alternative option, such as composting or anaerobic digestion, and is then prevented from being disposed of in a SWDS. The methane is generated from waste disposed or avoided from disposal during the crediting period. In these cases, the tool can be applied for both ex ante and ex post estimation of emissions. These project activities may apply the simplified approach detailed in 0 when calculating baseline emissions.	Not applicable.
TOOL32	The use of this methodological tool is not mandatory for the project participants of a CDM project activity or CDM POA for demonstrating their additionality.	Applicable.
	This methodological tool shall be applied in conjunction with a small-scale or large-scale methodology which refers to this tool.	Applicable. This tool is applied in conjunction with small-scale methodology AMS-III.G. (version 10.0) .

3.3 Project Boundary

The project boundary is the site of the project activity—Shangluo MSW treatment site, where the LFG is captured and used, and also includes all the power sources connected to the NCPG, which expands throughout Shaanxi Province, Gansu Province, Qinghai Province, Ningxia Autonomous Region, Xinjiang Autonomous Region.

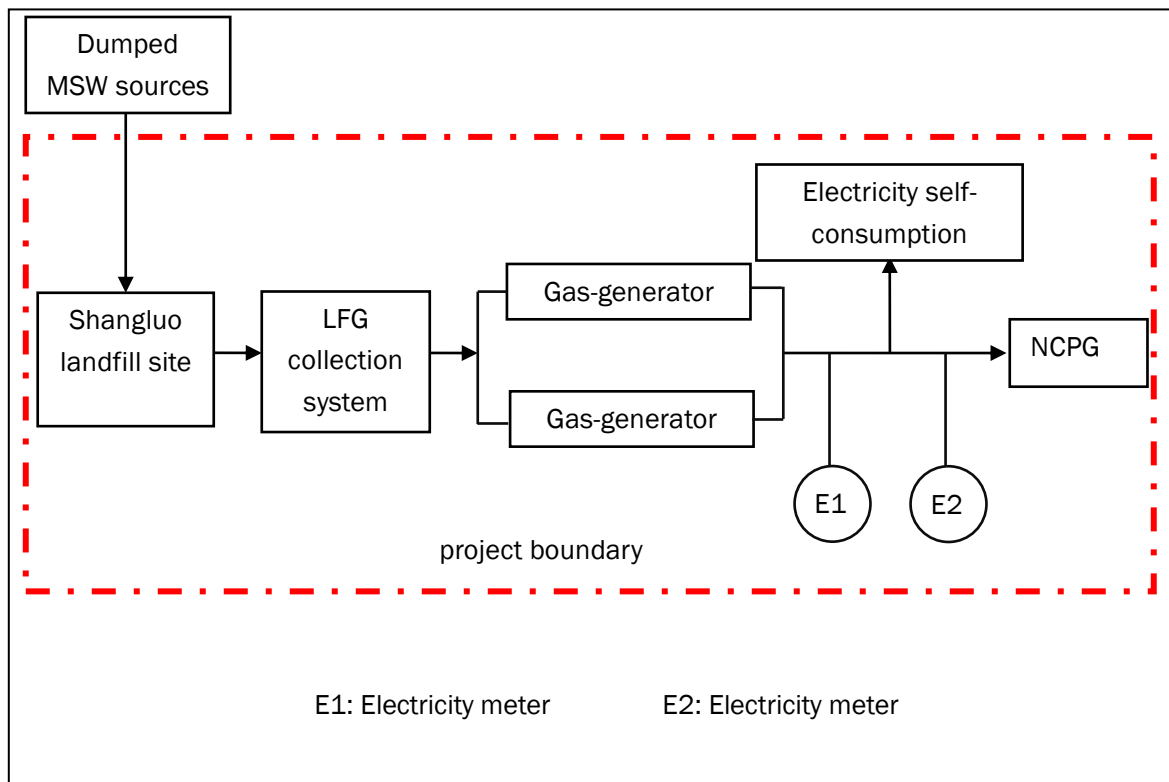


Figure 3.1 the Diagram of Project Boundary

Table 3.1 Summary of greenhouse gases and sources included in and excluded from the project boundary

Source	Gas	Included?	Justification/Explanation
Baseline	CH ₄	Yes	The major source of emissions in the baseline.
	N ₂ O	No	N ₂ O emissions are small compared to CH ₄ emissions from landfills. Exclusion of this gas is conservative.
	CO ₂	No	CO ₂ emissions from the decomposition of organic waste are not accounted since the CO ₂ is also released under the project activity.

Source		Gas	Included?	Justification/Explanation
	Emission from electricity generation	CO ₂	Yes	The major source of emissions in the baseline.
		CH ₄	No	Excluded for simplification, this is conservative.
		N ₂ O	No	Excluded for simplification, this is conservative.
	Emissions from heat generation	CO ₂	No	No heat generation is included in the project.
		CH ₄	No	No thermal energy generation is included in the project.
		N ₂ O	No	No thermal energy generation is included in the project.
	Emissions from the use of natural gas	CO ₂	No	Excluded for simplification. This is conservative
		CH ₄	No	No supply of LFG is included in the project.
		N ₂ O	No	Excluded for simplification. This is conservative
Project activity	Emissions from fossil fuel consumption for purposes other than electricity generation or transportation due to the project activity	CO ₂	No	Excluded because there is no fossil fuel consumption.
		CH ₄	No	Excluded because there is no fossil fuel consumption.
		N ₂ O	No	Excluded because there is no fossil fuel consumption.
	Emissions from electricity consumption due to the project activity	CO ₂	Yes	Main emission source.
		CH ₄	No	Excluded for simplification. This emission source is very small compared to CO ₂ emissions.
		N ₂ O	No	Excluded for simplification. This emission source is very small compared to CO ₂ emissions.
	Emissions from flaring	CO ₂	No	Excluded because there is no flaring in the project.
		CH ₄	No	Excluded because there is no flaring in the project

Source	Gas	Included?	Justification/Explanation
Emissions from distribution of LFG using trucks and dedicated pipelines	N ₂ O	No	Excluded because there is no flaring in the project
	CO ₂	No	No supply of LFG is included in the project.
	CH ₄	No	No supply of LFG is included in the project.
	N ₂ O	No	No supply of LFG is included in the project.

3.4 Baseline Scenario

According to the section 5.3 of the methodology AMS-III.G (version 10.0), Section 5.2 of methodology AMS-I.D (Version 18.0), the project chooses simplified procedures to identify the baseline scenario.

The LFG from Shangluo MSW treatment site is vented to the atmosphere prior the implementation of the project for safety concerns, and the majority electricity generated by the LFG will be exported to the Northwest China Power Grid (NCPG), and a little of which is consumed by the project.

LFG from Shangluo MSW treatment site is emitted to the atmosphere directly.

Equivalent electricity generated by the project is supplied by the Northwest China Power Grid (NCPG), which is dominated by fossil fuel based power plants.

3.5 Additionality

The project uses the following steps to demonstrate additionality. The project meets the additionality criteria as outlined in the VCS methodology as:

Step 1: Regulatory Surplus. the project is not mandated by any legal requirements. Please refer to section 3.5.1 below for details

Step 2: Positive List. The project conforms to the positive list as stipulated in the applies the simplified procedures in section 5.3 of the methodology AMS-III.G (version 10.0), Section 5.2 of methodology AMS-I.D (Version 18.0) and CDM TOOL32- "Positive lists of technologies " (version 04.0). Please refer to section 3.5.2 below for details.

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country

☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes

☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes

☒ No

According to the relevant applicable laws and regulations at the time of submission of registration, i.e. Renewable Energy Law of the Peoples Republic of China which came into effect on 01-Jan-2006, Agenda for China's 14th Five-year Plan (2021-2025), there are no new national and / or sectoral policies that could affect the baseline scenario during the crediting period. Therefore, the project is surplus to regulatory requirements

3.5.2 Additionality Methods

According to the section 5.3 of the methodology AMS-III.G (version 10.0), section 5.2 of methodology AMS-I.D (Version 18.0) and CDM TOOL32 (Version 4.0), project participants may either apply the simplified procedures in section 5.3.1 below or the procedures in section 5.3.2 to select the most plausible baseline scenario and demonstrate additionality. The project chooses simplified procedures to demonstrate additionality. Tool32 (Positive lists of technologies) is used to demonstrate additionality.

Requirements of tool 32 (Positive lists of technologies)	The project
5.1Waste handling and disposal 5.1.1Landfill gas recovery and its gainful use The project activities and PoAs at new or existing landfills (greenfield or brownfield)are deemed automatically additional if it is demonstrated that prior to the implementation of the project activities and PoAs the landfill gas(LFG)was only vented and or flared (in the case of brownfield projects)or would have been only vented and/or flared(in the case of greenfield projects)but not utilized for energy generation, and that under the project activities and PoAs any of the following conditions are met: (a) The LFG is used to generate electricity in one or several power plants with a total nameplate capacity that equals or is below 10 MW; (b) The LFG is used to generate heat for internal or external consumption	The LFG from Shangluo MSW treatment site is vented to the atmosphere prior the implementation of the project for safety concerns. Total nameplate capacity of the project is 1MW, which is below 10MW.

(c)The LFG is flared	
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Conclusion: the project is additional.

3.6 Methodology Deviations

No methodology deviations are applied for the project.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The project utilizes the LFG for generation to substitute the equivalent electricity supplied by the grid, resulting in CH₄ and CO₂ emissions, which will be calculated as follows according to the methodology AMS-III.G. and AMS-I.D.

Baseline emissions associated with the SWDS:

$$BE_{CH_4,y} = \eta_{PJ} \times BE_{CH_4,SWDS,y} - (1 - OX) \times F_{CH_4,BL,y} \times GWP_{CH_4} \quad \text{Equation (1)}$$

Where:

- $BE_{CH_4,y}$ = Baseline emissions associated with the SWDS in year y (tCO₂e/yr)
- η_{PJ} = Efficiency of the LFG capture system that will be installed in the project activity. It is used for ex ante estimation only. A default value of 50 per cent may be used
- $BE_{CH_4,SWDS,y}$ = Methane emission potential of a solid waste disposal site (in tCO₂e)
- OX = Oxidation factor (reflecting the amount of methane from SWDS that is oxidised in the soil or other material covering waste) (dimensionless). A default value of 0.1 may be used
- $F_{CH_4,BL,y}$ = Methane emissions that would be captured and destroyed to comply with national or local safety requirement or legal regulations in the year y (t CH₄). The relevant procedures in “ACM0001: Flaring or use of landfill gas” may be followed, as well as taking into account the compliance with the relevant local laws and regulation if such laws and regulations exist

GWP_{CH_4} = Global Warming Potential for methane

Ex ante determination of $BE_{CH_4,SWDS,y}$

According to the AMS-III.G., $BE_{CH_4,SWDS,y}$ is calculated using the methodological tool "Emission from solid waste disposal sites", which is calculated as follows:

$$BE_{CH_4,SWDS,y} = \varphi_y \times (1 - f_y) \times GWP_{CH_4} \times (1 - OX) \times \frac{16}{12} \times F \times DOC_{f,y} \times MCF_y \times \sum_{x=1}^y \sum_j (W_{j,x} \times DOC_j \times e^{-k_j \times (y-x)} \times (1 - e^{k_j})) \quad \text{Equation (2)}$$

Where:

$BE_{CH_4,SWDS,y}$	=	Baseline methane emissions occurring in the year y
φ_y	=	Model correction factor to account for model uncertainties for year y
f_y	=	Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emission of methane to the atmosphere in year y
GWP_{CH_4}	=	Global Warming Potential of methane
OX	=	Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)
F	=	Fraction of the methane in the SWDS gas (volume fraction)
$DOC_{f,y}$	=	Fraction of degradable organic carbon (DOC) that decomposes under the specific conditions occurring in the SWDS for year y (weight fraction)
MCF_y	=	Methane correction factor for year y
$W_{j,x}$	=	Amount of organic waste type j disposed/prevented from disposal in the SWDS in the year y (t)
DOC_j	=	Fraction of degradable organic carbon in the waste type j (weight fraction for year y)
k	=	Decay rate for the waste type (1/yr)
j	=	Type of residual waste or types of waste in the MSW
x	=	Years of the crediting period for which waste is disposed at the SWDS, extending from the first year in the time period ($x=1$) to year y ($x=y$)
y	=	Years of the crediting period for which methane emissions are calculated

Determination of $F_{CH_4, BL, y}$

This section provides a procedure to determine the amount of methane that would have been captured and destroyed (by flaring) in the baseline due to regulatory or contractual requirements, to address safety and odour concerns, or for other reasons (collectively referred to as requirement in this section). The four cases in **Table 4.1** are distinguished. The appropriate case should be identified, and the corresponding instructions followed.

Table 4.1.Cases for determining methane captured and destroyed in the baseline

Situation at the start of the project activity	Requirement to destroy methane	Existing LFG capture and destruction system
Case 1	No	No
Case 2	Yes	No
Case 3	No	Yes
Case 4	Yes	Yes

Currently China has regulations in place to deal with the management of landfills and to encourage utilization of LFG. Those regulations are:

“Standard for Pollution Control on the Landfill Site of Municipal Solid Waste” (GB 16889-2008), which became effective in 2008, issued by the Environment Protection Administration.

“Technical Code for Municipal Solid Waste Sanitary Landfill” (GB 50869-2013), issued by the Ministry of Construction in 2013.

According to item 5.15 of GB16889-2008, if the designed landfill capacity is more than 2.5 million tons and the landfill thickness is more than 20m, methane utilization facilities or flare burning facilities shall be built to treat the landfill gas containing methane. For municipal solid waste landfills smaller than the above scale, technologies that can effectively reduce methane generation and emission shall be adopted or flare combustion facilities shall be used to treat methane containing landfill gas.

Item 11.1.1 of GB 50869-2013 stipulates that the landfill site must be equipped with effective landfill gas drainage facilities to prevent the natural accumulation and migration of landfill gas, causing fire and explosion. Item 11.1.3 stipulates that if the landfill does not have the conditions for landfill gas utilization, the flare method shall be adopted for combustion treatment, and the process that can effectively reduce the generation and emission of methane shall be adopted. The old landfills that are not safe and stable should be equipped with effective landfill gas drainage facilities. Among them, item 11.1.1 is mandatory and must be strictly implemented.

In fact, the LFG of Shangluo landfill is emitted to atmosphere without LFG capture system prior the implementation of the project. Therefore, Case 2 listed in the table 4.1 is applicable for the project.

The requirements above don't specify any amount or percentage of LFG that should be destroyed. In this situation:

$$F_{CH_4,BL,y} = F_{CH_4,BL,R,y} \quad \text{Equation (3)}$$

$$F_{CH_4,BL,R,y} = 0.2 \times F_{CH_4,PJ,capt,y} \quad \text{Equation (4)}$$

Where:

$F_{CH_4,BL,R,y}$ = Amount of methane in the LFG which is flared in the baseline due to a requirement in year y (tCH₄/yr)

$F_{CH_4,PJ,capt,y}$ = Amount of methane in the LFG which is captured in the project activity in year y (tCH₄/yr)

Baseline emissions associated with electricity generation ($BE_{EC,y}$)

$$BE_{EC,y} = EG_{PJ,y} \times EF_{grid,y} \quad \text{Equation (5)}$$

$BE_{EC,y}$ = Baseline emissions associated with electricity generation in year y (t CO₂)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)

$EF_{grid,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (t CO₂/MWh)

Calculation of $EG_{PJ,y}$

The project is the installation of a greenfield power plant, therefore:

$$EG_{PJ,y} = EG_{PJ,facility,y} \quad \text{Equation (6)}$$

Where:

$EG_{PJ,facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh)

According to AMS I.D, the emission factor shall be calculated in a transparent and conservative manner as follows:

(a) A combined margin (CM), consisting of the combination of operating margin (OM) and build margin (BM) according to the procedures prescribed in the “Tool to calculate the emission factor for an electricity system”; or

(b) The weighted average emissions (in t CO₂/MWh) of the current generation mix. The data of the year in which project generation occurs must be used.

According to “Tool to calculate the emission factor for an electricity system” (version 07.0). The following six steps are applied to determine OM, BM, and CM used for calculating the project

baseline emissions:

- Step 1:** Identify the relevant electricity systems;
- Step 2:** Choose whether to include off-grid power plants in the project
- Step 3:** Select a method to determine the operating margin (OM);
- Step 4:** Calculate the operating margin emission factor according to the
- Step 5:** Calculate the build margin (BM) emission factor;
- Step 6:** Calculate the combined margin (CM) emission factor.

Step 1: Identity the relevant electricity systems

According to the Tool to calculate the emission factor for an electricity system (Version 07.0), project participants may delineate the project electricity system using any of the following options:

Option 1. A delineation of the project electricity system and connected electricity systems published by the DNA or the group of the DNAs of the host country(ies), In case a delineation is provided by a group of DNAs, the same delineation should be used by all the project participants applying the tool in these countries;

Option 2. A delineation of the project electricity system defined by the dispatch area of the dispatch centre responsible for scheduling and dispatching electricity generated by the project activity. Where the dispatch area is controlled by more than one dispatch centre, i.e. layered dispatch area, the higher level area shall be used as a delineation of the project electricity system (e.g. where regional dispatch centres are required to comply with dispatch orders of the national dispatch centre then area controlled by the national dispatch centre shall be used);

Option 3. A delineation of the project electricity system defined by more than one independent dispatch areas, e.g. multi-national power pools.

The Chinese DNA has published a delineation of the project electricity system and connected electricity systems, Option 1 is applied for the project. According to the delineations, the Northwest China Power Grid (NCPG) is identified as the relevant electric power system of the project, which includes Shaanxi Province, Gansu Province, Qinghai Province, Ningxia Autonomous Region, Xinjiang Autonomous Region Power Grids.

Step 2. Choose whether to include off-grid power plants in the project electricity system (optional).

Option I: Only grid power plants are included in the calculation.

Option II: Both grid power plants and off-grid power plants are included in the calculation.

Following the calculation of the Ministry of Ecology and Environment of the People's Republic of China, and statistical data is available, Option I is chosen.

Step 3. Select a method to determine the operating margin (OM)

“Tool to calculate the emission factor for an electricity system (Version 7.0)” offers four methods for the calculation of the operating margin emission factor(s) ($EF_{grid,OM,y}$):

- (a) Simple OM; or
- (b) Simple adjusted OM; or
- (c) Dispatch data analysis OM; or
- (d) Average OM.

Method (a) -Simple OM is chosen for calculation and low-cost/must-run resources constitute less than 50% of the total grid generation in average of the five most recent years⁸.

For simple OM, the emission factor can be calculated using either of the two following data vintages:

- (a) Ex ante option: If the ex ante option is chosen, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the mission factor during the crediting period is required. For grid power plants, use a 3-year generation-weighted average, based on the most recent data available at the time of submission of the CDM-PDD to the DOE for validation. For off-grid power plants, use a single calendar year within the 5 most recent calendar years prior to the time of submission of the CDM-PDD for validation;
- (b) Ex post option: If the ex post option is chosen, the emission factor is determined for the year in which the project activity displaces grid electricity, requiring the emissions factor to be updated annually during monitoring. If the data required to calculate the emission factor for year y is usually only available later than six months after the end of year y, alternatively the emission factor of the previous year y-1 may be used. If the data is usually only available 18 months after the end of year y, the emission factor of the year proceeding the previous year y-2 may be used. The same data vintage (y, y-1 or y-2) should be used throughout all crediting periods

Project participant employs ex ante option for its operation margin calculation.

Step 4. Calculate the operating margin emission factor according to the selected method

The simple OM emission factor is calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost / must-run power plants / units.

The simple OM may be calculated by one of the following two options:

- (a) Option A: Based on the net electricity generation and a CO₂ emission factor of each power unit; or

⁸ 2023 China baseline emission factor of regional power grid.

- (b) Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system. Option B can only be used if:
- (i) The necessary data for Option A is not available: and
 - (ii) Only nuclear and renewable power generation are considered as low-cost/must-run power sources and the quantity of electricity supplied to the grid by these sources is known; and
 - (iii) Off-grid power plants are not included in the calculation (i.e., if Option I has been chosen in Step 2).

Since the data of each power plant/unit is unavailable, Option A is not applicable to the project. The project adopts Option B to calculate the operating margin emission factor ($EF_{grid,OM,y}$) of NCPG.

$$EF_{grid,OMsimple,y} = \frac{\sum_i FC_{i,y} \times NCV_{i,y} \times EF_{CO_2,i,y}}{EG_y} \quad \text{Equation(7)}$$

Where:

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO ₂ emission factor in year y (tCO ₂ /MWh)
$FC_{i,y}$	=	Amount of fuel type i consumed in the project electricity system in year y (mass or volume unit)
$NCV_{i,y}$	=	Net calorific value (energy content) of fuel type i in year y (GJ/mass or volume unit)
$EF_{CO_2,i,y}$	=	CO ₂ emission factor of fossil fuel type i in year y (tCO ₂ e/GJ), 2006 IPCC Guidelines for default values
EG_y	=	Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must run power plants/units in year y (MWh)
i	=	All fuel types combusted in power sources in the project electricity system in year y
y	=	The relevant year as per the data vintage chosen in Step 3

For Central China Power Grid and Southern Power Grid, use the adjusted simple OM method to calculate their OM emission factor. The adjusted simple OM divides the generator sets into two categories: low cost/must run power plants and all the other power plants except low cost/ must run power plants. According to the weighted average of each unit's annual power supply emission factor, the probability weights of the two units participating in generation scheduling (λ and $1-\lambda$) are obtained. Finally, the unit power supply emission factor of the two types of generating units is adjusted according to the participation in generation. The probability weights of degrees are

weighted and averaged to obtain the adjusted simple OM. The specific calculation formula is as follows:

$$EF_{grid,OM-adj,y} = (1 - \lambda_y) \times EF_{grid,OMsimple,y} + \lambda_y \times \frac{\sum_k EG_{k,y} \times EF_{EL,k,y}}{\sum_k EG_{k,y}} \quad \text{Equation(8)}$$

Where:

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO ₂ emission factor in year y (tCO ₂ /MWh), according to the equation (6)
$EG_{k,y}$	=	Net electricity generated and delivered to the grid by low-cost/must run power plants/units type k in year y (MWh)
$EF_{EL,k,y}$	=	CO ₂ emission factor by low-cost/must run power plants/units type k in year y (tCO ₂ /MWh)
k	=	Type of low-cost/must run power plants/units in year y
λ_y	=	Probability weight of low-cost/must run power plants/units participating in generation scheduling in year y
y	=	The relevant year as per the data vintage chosen in Step 3

If available, NCV_{i,y} and EF_{CO2,i,y} from the fuel supplier of the power plants in invoices may be used; or regional or national average default values may be used. In this PD, NCV_{i,y} of different fuels are obtained from China Energy Statistical Yearbook 2018. Emission factors (EF_{CO2,i,y}) of each type of fossil fuel come from 2019 refinement to 2006 IPCC default values.

The Simple OM Emission Factor ($EF_{grid,OMsimple,y}$) of the project is calculated based on the fuel consumption data for electricity generation of the China Southern Power Grid, not including those of low-operating cost and must-run power plants, such as wind power, hydropower and nuclear etc. These data are obtained from the China Electric Power Yearbook (2020~2022, published annually) and China Energy Statistical Yearbook (2017~2019). Based on these data, the Simple OM Emission Factor ($EF_{grid,OMsimple,y}$) of the Northwest China Power Grid is calculated as 0.9014 tCO₂/MWh. Details of the calculations and the published data from the Chinese DNA⁹, which uses official national statistics.

Step 5. Calculate the build margin emission factor

In terms of vintage of data, project participants can choose between one of the following two options:

- (a) Option 1: For the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of CDM PDD submission to the DOE for validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information

⁹ <https://ccer.cets.org.cn/notice/noticeDetail?bulletinInfold=1175122354980917248>

available on units already built at the time of submission of the request for renewal of the crediting period to the DOE. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period.

- (b) Option 2: For the first crediting period, the build margin emission factor shall be updated annually, ex-post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex ante, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

The project applies option 1 to calculate the build margin emission factor ex-ante.

According to the “Tool to calculate the emission factor for an electricity system”, the sample group of power units used to calculate the build margin should be determined as per the following procedure, consistent with the data vintage selected above:

- (a) Identify the set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently ($SET_{5 \text{ units}}$) and determine their annual electricity generation ($AEG_{SET-5-units}$, in MWh);
- (b) Determine the annual electricity generation of the project electricity system, excluding power units registered as CDM project activities (AEG_{total} , in MWh). Identify the set of power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently and that comprise 20% of AEG_{total} (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) ($SET_{\geq 20\%}$) and determine their annual electricity generation ($AEG_{SET \geq 20\%}$, in MWh);
- (c) From $SET_{5-units}$ and $SET_{\geq 20 \text{ percent}}$ select the set of power units that comprises the larger annual electricity generation (SET_{sample});

Identify the date when the power units in SET_{sample} started to supply electricity to the grid. If none of the power units in SET_{sample} started to supply electricity to the grid more than 10 years ago, then use SET_{sample} to calculate the build margin. Ignore steps (d), (e) and (f).

Otherwise:

- (d) Exclude from SET_{sample} the power units which started to supply electricity to the grid more than 10 years ago. Include in that set the power units registered as CDM project activity, starting with power units that started to supply electricity to the grid most recently, until the electricity generation of the new set comprises 20% of the annual electricity generation of the project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) to the extent is possible. Determine for the resulting set ($SET_{sample-CDM}$) the annual electricity generation ($AEG_{SET-sample-CDM}$, in MWh);

If the annual electricity generation of that set is comprises at least 20% of the annual electricity generation of the project electricity system (i.e. $AEG_{SET-sample-CDM} \geq 0.2 \times AEG_{total}$), then use the sample group $SET_{sample-CDM}$ to calculate the build margin. Ignore steps (e) and (f).

- (e) Include in the sample group $SET_{sample-CDM}$ the power units that started to supply electricity to the grid more than 10 years ago until the electricity generation of the new set comprises 20% of the annual electricity generation of the project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation);
- (f) The sample group of power units m used to calculate the build margin is the resulting set ($SET_{sampleCDM} > 10yrs$)

The build margin emissions factor is the generation-weighted average emission factor (tCO_2/MWh) of all power units m during the most recent year y for which power generation data is available, calculated as follows:

$$EF_{grid,BM,y} = \frac{\sum_m EG_{m,y} \times EF_{EL,m,y}}{\sum_m EG_{m,y}} \quad \text{Equation (9)}$$

Where:

- $EF_{grid,BM,y}$ = Build margin CO_2 emission factor in year y ($t CO_2/MWh$)
- $EG_{m,y}$ = Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
- $EF_{EL,m,y}$ = CO_2 emission factor of power unit m in year y ($t CO_2/MWh$)
- m = Power units included in the build margin
- y = Most recent historical year for which electricity generation data is available

Under the current circumstances in China, the power plants consider the Build Margin data as important business data and will not have them published. Therefore, it is difficult to obtain the data of five power plants that have been put into operation most recently or the newly installed power plant capacity additions in the electricity system that comprise 20% of the system generation.

According to the instructions of China DNA, for the determination of the set of samples, a sample merging processing in some degree has been adopted due to that the power generation data, energy consumption data or thermal efficiency data of each plant cannot be consulted in the public statistical data. In this calculation, the newly-installed power units in the past years are classified by year, province and power generation technology, and the same type of newly installed power units in the same province and in the same year are bundled as a "newly-installed power units".

The power generation of each "newly-installed power units" in the most recent year y is estimated based on its installed capacity and the number of power generation utilization hours in year y . The formula is as follows:

$$EG_{m,y} = CAP_m \times H_{m,y} \quad \text{Equation(10)}$$

Where:

$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
CAP_m	=	Installed capacity of electricity generated and delivered to the grid by power unit m in year y (MW)
$H_{m,y}$	=	The number of power utilization hours (h) of electricity generated and delivered to the grid by power unit m in year
y	=	And it selects the average utilization hours of similar units in the province in which it is located in year y ;
m	=	The sample group of power units

The power unit m is selected from the "newly-installed power plants" in the most recent year y (For the calculation of the grid BM in 2023, the y is equal to 2021) to the "newly- installed power plants" in the earlier year, until the cumulative power generation reaches 20% of the total power generation in the year y .

Since the newly-installed power units of the same type (k) in the same province (A) and the same year (t) are bundled into the "newly-installed power units", the CAP_m is equal to the statistical data of recent installed capacity of a given unit type(k) in a given year(t) in a given province (A).

$$CAP_m = CAP_m|_{m=(A,t,k)} = CAP_{A,t,k} \quad \text{Equation(11)}$$

Where:

CAP_m	=	Installed capacity of electricity generated and delivered to the grid by power unit m in year y (MW), and m is equivalent to an established combination of (A , t , k)
$CAP_{A,t,k}$	=	Capacity of newly-installed power units of a given province (A), given year (t), and given unit type (k) (MW)
A	=	It is the various provincial regions covered by the regional power grid
t	=	It is the sampling year of the "newly-installed power units". For the calculation of the grid BM in 2023, t is equal to 2021,2020, 2018, until the units that comprise at least 20 percent of the system generation in 2021

k = It is the power generation technology classification of "newly-installed power units", which is divided into: hydro power, coal-thermal power, gas-thermal power, oil-thermal power, Waste-thermal power plant, other thermal power¹⁰nuclear power, wind power, solar power, and others¹¹

As per tool, the CO₂ emission factor of each power unit m ($EF_{EL,m,y}$) should be determined as per the tool in Step 4 section 6.4.1 for the simple OM, using Options A1, A2 or A3, using for y the most recent historical year for which electricity generation data is available, and using from the power units included in the build margin.

Because current statistics data cannot separate each power plant, for a power unit m, only data on electricity generation and the fuel types used is available. So, the option A2 is selected, the emission factor should be determined based on the CO₂ emission factor of the fuel type used and the efficiency of the power unit, as follows:

$$EF_{EL,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{m,y}} \quad \text{Equation (12)}$$

Where:

$EF_{EL,m,y}$ = CO₂ emission factor of power unit m in year y (tCO₂/MWh)

$EF_{CO2,m,i,y}$ = Average CO₂ emission factor of fuel type i used in power unit m in year y (tCO₂/GJ)

$\eta_{m,y}$ = Average net energy conversion efficiency of power unit m in year y (dimensionless)

3.6 = Conversion coefficient of thermal work equivalent of electricity (GJ/MWh)

According to Equation (11), the unit electricity emission factor of the hydro-power, nuclear power, wind power, solar power, other thermal power, and others power generation technology in the "newly-installed power units" samples are is zero. The emission factor per unit of electricity for power generation from coal, gas, oil and electricity waste power is calculated based on Equation (11). Since the average net energy conversion efficiency of each sample ($\eta_{m,y}$) cannot be obtained, the power supply thermal efficiency of the best commercialized technology of coal, gas, oil and waste power ($\eta_{Best,m,y}$) is better than $\eta_{m,y}$. It is conservative to use $\eta_{Best,m,y}$ for the calculation of $EF_{EL,m,y}$. Therefore, the formula is as follows:

$$EF_{Best,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{Best,m,y}} \quad \text{Equation (13)}$$

¹⁰ refers to waste heat and pressure, straw, bagasse, forest wood power generation

¹¹ Refers to power generation such as geothermal energy and ocean energy.

Where:

$EF_{Best,m,y}$	=	Emission factor of power unit m with the best technology commercially available in year y (tCO ₂ /MWh)
$\eta_{Best,m,y}$	=	Power generation efficiency of the best technology commercially available in year y (dimensionless)
m	=	Sample of "newly-installed power units" generated by coal, gas, oil and waste incineration in a province in a certain year

The emission factors of coal, natural gas, fuel oil and municipal solid waste (MSW) are sourced from 2006 IPCC Guidelines for National Greenhouse Gas Inventories, and the lower values of the 95% confidence interval are applied to be conservative.

No fuel oil power units were constructed in 2021. The power generation efficiency of the best technology commercially available for fuel oil units comes from Baseline Emission Factors for Regional Grids in China of previous years, i.e., 52.9%.

According to the above steps, the build margin emission factor, $EF_{grid,BM,y}$ of the NCPG is calculated to be 0.3597 tCO₂/MWh, which is confirmed by 2023 Baseline Emission Factors for Regional Power Grids in China published by Ministry of Ecology and Environment of the People's Republic of China¹².

Step 6. Calculate the combined margin emission factor

Combined Margin emission factor($EF_{grid,CM,y}$) is calculated as the weighted average of the operating margin emission factor($EF_{grid,OM,y}$) and the build margin emission factor ($EF_{grid,BM,y}$),where the weights ω_{OM} and ω_{BM} ,by default, are 0.5 and 0.5 in the crediting period, and $EF_{grid,OM,y}$ and $EF_{grid,BM,y}$ are calculated as described above and are expressed in tCO₂/MWh.

$$EF_{grid,CM,y} = \omega_{OM} \times EF_{grid,OM,y} + \omega_{BM} \times EF_{grid,BM,y} \quad \text{Equation(14)}$$

$$EF_{grid,CM,y} = 0.5 \times 0.9014 + 0.5 \times 0.3597 = 0.63055 (\text{tCO}_2/\text{MWh})$$

The $EF_{grid,OM,y}$, $EF_{grid,BM,y}$ and $EF_{grid,CM,y}$ are ex-ante calculation and are fixed during the credit period.

Table 4.2: Necessary values to calculate the baseline emissions associated with the SWDS

Parameter	Value	Data source
OX	0.1	Using IPCC 2006 Guidelines default value

¹² <https://ccer.cets.org.cn/notice/noticeDetail?bulletinInfolId=1175122354980917248>

GWP_{CH_4}	28	IPCC Fifth Assessment Report (AR5)	
η_{PJ}	80%	FSR of the project	
ϕ_y	0.75	“Emissions from solid waste disposal sites” (version 08.1)	
OX_{top_layer}	0.1	Emissions from solid waste disposal sites” (version 08.1)	
f_y	0	“Emissions from solid waste disposal sites” (version 08.1)	
F	0.5	IPCC 2006 Guidelines for National Greenhouse Gas Inventories	
$DOC_{f,y}$	0.5	IPCC 2006 Guidelines for National Greenhouse Gas Inventories	
MCF_y	1.0	IPCC 2006 Guidelines for National Greenhouse Gas Inventories	
$W_{j,x}$	Annual landfilled MSW (tons)		The parameter is calculated as total waste amount dumped in the landfill site in the year x multiplied by organic waste type j fraction on wet basis. Both total waste amount and waste
	Year	Annual landfilled MSW (tons)	
	2008	44,275	
	2009	60,225	
	2010	61,320	
	2011	62,415	
	2012	63,875	
	2013	64,970	

	2014	66,795	type j fraction are referred to FSR.									
	2015	68,620										
	2016	70,445										
	2017	72,270										
	2018	74,095										
	2019	76,650										
	2020	78,840										
	2021	79,205										
	2022	82,125										
	2023	85,045										
	2024	87,965										
	2025	90,885										
	2026	94,535										
	2027	98,550										
	2028	102,930										
	2029	107,675										
	2030	112,420										
	2031	117,530										
	2032	122,640										
	Organic waste type j fraction on wet basis:											
<table><tr><td>Waste type j</td><td>Weight Fraction (wet waste)</td></tr><tr><td>W₁-Wood and wood products</td><td>0.0301</td></tr><tr><td>W₂-Pulp, paper and cardboard</td><td>0.0834</td></tr><tr><td>W₃-Food, food waste, beverages and tobacco</td><td>0.4027</td></tr><tr><td>W₄-Textiles</td><td>0.0304</td></tr></table>			Waste type j	Weight Fraction (wet waste)	W ₁ -Wood and wood products	0.0301	W ₂ -Pulp, paper and cardboard	0.0834	W ₃ -Food, food waste, beverages and tobacco	0.4027	W ₄ -Textiles	0.0304
Waste type j	Weight Fraction (wet waste)											
W ₁ -Wood and wood products	0.0301											
W ₂ -Pulp, paper and cardboard	0.0834											
W ₃ -Food, food waste, beverages and tobacco	0.4027											
W ₄ -Textiles	0.0304											

	W ₅ -Garden, yard and park waste	0.1341		
	W ₆ -Glass, plastic, metal other inert	0.3193		
DOC _j	Waste type j	DOC _j (% wet waste)		“Emissions from solid waste disposal sites” (version 08.1)
	Wood and wood products	43		
	Pulp, paper and cardboard (other than sludge)	40		
	Textiles	24		
	Garden, yard and park waste	20		
	Glass, plastic, metal, other inert waste	0		
k _j	Waste type j		k _j (MAT ≤ 20°C, MAP/PET < 1)	“Emissions from solid waste disposal sites” (version 08.1)
	Slowly degrading	Pulp, paper, cardboard (other than sludge)	0.04	
		Wood, wood products and straw	0.06	
	Moderately degrading	Other(non-food) organica putrescible garden and park waste	0.04	
	Rapidly degrading	Food, food waste, sewage sludge, beverages and tobacco	0.05	

Table 4.3 The estimation of baseline emissions associated with the SWDS

Year	η _{PJ}	BE _{CH4,SWDS,y}	GWP _{CH4}	OX _{top_layer}	F _{CH4,BL,y}	BE _{CH4,y}
23-Mar-2023 to 31-Dec-2023	80%	26,264	28	0.1	150	17,229
01-Jan-2024 to 31-Dec-2024	80%	35,776	28	0.1	204	23,469
01-Jan-2025 to 31-Dec-2025	80%	37,821	28	0.1	216	24,810
01-Jan-2026 to 31-Dec-2026	80%	39,919	28	0.1	228	26,187
01-Jan-2027 to 31-Dec-2027	80%	42,083	28	0.1	240	27,606

01-Jan-2028 to 31-Dec-2028	80%	44,325	28	0.1	253	29,077
01-Jan-2029 to 31-Dec-2029	80%	46,657	28	0.1	267	30,607
01-Jan-2030 to 31-Dec-2030	80%	49,074	28	0.1	280	32,192
01-Jan-2031 to 31-Dec-2031	80%	51,587	28	0.1	295	33,841
01-Jan-2032 to 31-Dec-2032	80%	54,192	28	0.1	310	35,549
01-Jan-2033 to 22-Mar-2033	80%	11,313	28	0.1	65	7,421

Table 4.4 The estimation of baseline emissions associated with electricity generation

Year	EC _{BL,k,y}	EF _{grid,CM,y}	BE _{EC,y}
23-Mar-2023 to 31-Dec-2023	3,153	0.63055	1,987
01-Jan-2024 to 31-Dec-2024	4,294	0.63055	2,707
01-Jan-2025 to 31-Dec-2025	4,540	0.63055	2,862
01-Jan-2026 to 31-Dec-2026	4,792	0.63055	3,021
01-Jan-2027 to 31-Dec-2027	5,051	0.63055	3,185
01-Jan-2028 to 31-Dec-2028	5,320	0.63055	3,354
01-Jan-2029 to 31-Dec -2029	5,600	0.63055	3,531
01-Jan-2030 to 31-Dec-2030	5,890	0.63055	3,714
01-Jan-2031 to 31-Dec-2031	6,192	0.63055	3,904
01-Jan-2032 to 31-Dec-2032	6,505	0.63055	4,101
01-Jan-2033 to 22-Mar-2033	1,358	0.63055	856

Table 4.5 The estimation of baseline emissions

Year	BE _{CH4,y}	BE _{EC,y}	BE _y
23-Mar-2023 to 31-Dec-2023	17,229	1,987	19,216
01-Jan-2024 to 31-Dec-2024	23,469	2,707	26,176
01-Jan-2025 to 31-Dec-2025	24,810	2,862	27,672
01-Jan-2026 to 31-Dec-2026	26,187	3,021	29,208
01-Jan-2027 to 31-Dec-2027	27,606	3,185	30,791
01-Jan-2028 to 31-Dec-2028	29,077	3,354	32,431
01-Jan-2029 to 31-Dec -2029	30,607	3,531	34,138
01-Jan-2030 to 31-Dec-2030	32,192	3,714	35,906
01-Jan-2031 to 31-Dec-2031	33,841	3,904	37,745
01-Jan-2032 to 31-Dec-2032	35,549	4,101	39,650
01-Jan-2033 to 22-Mar-2033	7,421	856	8,277

4.2 Project Emissions

As per AMS I.D (version 18.0), project emission of the project is zero.

As per AMS III.G (version 10.0), project emission is calculated as per equation below:

$$PE_y = PE_{power,y} + PE_{flare,y} + PE_{process,y} \quad \text{Equation (15)}$$

Where:

PE_y	=	Project emission in year y (t CO ₂ e)
$PE_{power,y}$	=	Emission from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (t CO ₂ e)
$PE_{flare,y}$	=	Emissions from flaring or combustion of the landfill gas stream in the year y (t CO ₂ e)
$PE_{process,y}$	=	Emission from the landfill gas upgrading process in the year y (t CO ₂ e), determined by following the relevant procedures described in annex 1 of AMS-III.H.

The project does not involve the emissions from consumption of fossil fuels, only emission from the use of electricity for operation of the installed facilities is involved. The project will consume electricity delivered from the grid when the project is not in operation. And this part of electricity has already been deducted from quantity of electricity supplied by the project plant/unit to the grid when calculating the quantity of the net electricity generation supplied by the project plant/unit to the grid, which is then used to calculate the baseline emission associated with electricity generation ($BE_{EC,y}$), hence emissions from the use of electricity for the operation of the installed facilities is excluded, which means emission from the use of fossil fuel and electricity for the operation of the installed facilities equal to 0.

The project does not claim emission reductions from flaring, therefore, $PE_{flare,y}$ is equal to 0.

The project captures LFG for power generation and isn't involved upgrading process, therefore, $PE_{process,y}$ is equal to 0.

In conclusion, the project emission of the project activity is 0.

4.3 Leakage Emissions

There is no equipment transferred in the project, no leakage effects need to be accounted under AMS-III.G. and AMS-I.D.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

According to AMS-III.G., the emission reduction achieved by the project activity can be estimated ex-ante in the PD by:

$$ER_{y,estimated} = BE_y - PE_y - LE_y \quad \text{Equation (16)}$$

By combining all the above equations, the ex-ante estimate of the emission reduction is:

$$ER_{y,estimated} = \eta_{PJ} \times BE_{CH_4,SWDS,y} - (1 - OX) \times F_{CH_4,BL,y} \times GWP_{CH_4} + EG_{PJ,y} \times EF_{grid,y} - PE_y - LE_y \quad \text{Equation (17)}$$

Emission reductions(ex-post)

According to AMS-III.G., The actual emission reduction achieved by the project activity during the crediting period will be calculated using the amount of methane recovered and destroyed/gainfully used by the project activity, calculated as:

$$ER_{y,calculated} = (1 - OX) \times (F_{CH_4,PJ,y} - F_{CH_4,BL,y}) \times GWP_{CH_4} + EG_{PJ,y} \times EF_{grid,y} - PE_y - LE_y \quad \text{Equation (18)}$$

Where:

$$F_{CH_4,PJ,y} = \text{Methane captured and destroyed/gainfully used by the project activity in the year } y \text{ (t CH}_4\text{)}$$

The project utilizes the recovered methane for power generation. Therefore $F_{CH_4,PJ,y}$ is calculated as follows, based on the amount of monitored electricity generation without monitoring methane flow and concentration:

$$F_{CH_4,PJ,y} = \frac{EG_y \times 3600}{NCV_{CH_4} \times EE_y} \times D_{CH_4} \quad \text{Equation (19)}$$

Where:

EG_y	=	Electricity generation in year y(MWh)
3600	=	Conversion factor (1 MWh=3600 MJ)
NCV_{CH_4}	=	NCV of methane (MJ/NM ³) use default value:35.9 MJ/NM ³
EE_y	=	Energy Conversion Efficiency of the project equipment determined from one of the following options:
		- Specification provided by the equipment manufacturer specifically for biogas fuel only if the equipment is designed to utilize biogas as fuel if the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation - Default efficiency of 40 per cent

Therefore:

$$ER_{y,calculated} = (1 - OX) \times \left(\frac{EG_y \times 3600}{NCV_{CH_4} \times EE_y} \times D_{CH_4} - F_{CH_4,BL,y} \right) \times GWP_{CH_4} + EG_{PJ,y} \times EF_{grid,y} - PE_y - LE_y$$

Equation (20)

:

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
23-Mar-2023 to 31-Dec-2023	19,216	0	0	19,216	0	19,216
01-Jan-2024 to 31-Dec-2024	26,176	0	0	26,176	0	26,176
01-Jan-2025 to 31-Dec-2025	27,672	0	0	27,672	0	27,672
01-Jan-2026 to 31-Dec-2026	29,208	0	0	29,208	0	29,208
01-Jan-2027 to 31-Dec-2027	30,791	0	0	30,791	0	30,791
01-Jan-2028 to 31-Dec-2028	32,431	0	0	32,431	0	32,431
01-Jan-2029 to 31-Dec-2029	34,138	0	0	34,138	0	34,138
01-Jan-2030 to 31-Dec-2030	35,906	0	0	35,906	0	35,906
01-Jan-2031 to 31-Dec-2031	37,745	0	0	37,745	0	37,745
01-Jan-2032 to 31-Dec-2032	39,650	0	0	39,650	0	39,650
01-Jan-2033 to 22-Mar-2033	8,277	0	0	8,277	0	8,277

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	OX
Data unit	Dimensionless
Description	Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)
Source of data	Based on an extensive review of published literature on this subject, including the IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied	0.1
Justification of choice of data or description of measurement methods and procedures applied	Using IPCC 2006 Guidelines default value
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	GWP _{CH4}
Data unit	tCO ₂ e/t CH ₄
Description	Global warming potential of CH ₄
Source of data	IPCC Fifth Assessment Report (AR5)
Value applied	28
Justification of choice of data or description of measurement methods and procedures applied	Default value of 28 from IPCC Fifth Assessment Report (AR5). Shall be updated according to any future COP/CMP decisions
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	η_{PJ}
Data unit	Dimensionless

Description	Efficiency of the LFG capture system that will be installed in the project activity
Source of data	FSR of the project
Value applied	80%
Justification of choice of data or description of measurement methods and procedures applied	Using estimated value in FSR of the project
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	Φ_y
Data unit	-
Description	The model correction factor to account for model uncertainties
Source of data	Default value of the tool “Emissions from solid waste disposal sites” (version 08.1)
Value applied	0.75
Justification of choice of data or description of measurement methods and procedures applied	Application A is used to decide the value. And the value of dry conditions for used for the project since the ($MAT \leq 20^{\circ}C$, $MAP/PET < 1$).
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	f_y
Data unit	-
Description	Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y
Source of data	The tool “Emissions from solid waste disposal sites” (version 08.1)
Value applied	0

Justification of choice of data or description of measurement methods and procedures applied	It has been considered in the section 4.1, equation (1). Therefore, $f_y=0$
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	F
Data unit	-
Description	Fraction of methane in the SWDS gas (volume fraction)
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	Using IPCC 2006 Guidelines default value
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	DOC _{f,y}
Data unit	Weight fraction
Description	Default value for the fraction of degradable organic carbon (DOC) in MSW that decomposed in the SWDS
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	Using IPCC 2006 Guidelines default value
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	MCF_y
Data unit	-
Description	Methane correction factor
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied	1.0
Justification of choice of data or description of measurement methods and procedures applied	The project using application A to decide the value of MCF_y . The Shangluo MSW treatment site has controlled placement of waste, mechanical compacting and levelling of the waste, So MCF_y value of 1.0 for anaerobic managed solid waste disposal sites should be applied in line with "Emissions from solid waste disposal sites" (version 08.1)
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$W_{j,x}$																
Data unit	t																
Description	Amount of solid waste type j disposed or prevented from disposal in the SWDS in year x																
Source of data	Feasibility study report (FSR) of the project																
Value applied	Annual landfilled MSW (tons) <table border="1"> <thead> <tr> <th>Year</th><th>Annual landfilled MSW (tons)</th></tr> </thead> <tbody> <tr> <td>2008</td><td>44,275</td></tr> <tr> <td>2009</td><td>60,225</td></tr> <tr> <td>2010</td><td>61,320</td></tr> <tr> <td>2011</td><td>62,415</td></tr> <tr> <td>2012</td><td>63,875</td></tr> <tr> <td>2013</td><td>64,970</td></tr> <tr> <td>2014</td><td>66,795</td></tr> </tbody> </table>	Year	Annual landfilled MSW (tons)	2008	44,275	2009	60,225	2010	61,320	2011	62,415	2012	63,875	2013	64,970	2014	66,795
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2008	44,275																
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2014	66,795																

2015	68,620
2016	70,445
2017	72,270
2018	74,095
2019	76,650
2020	78,840
2021	79,205
2022	82,125
2023	85,045
2024	87,965
2025	90,885
2026	94,535
2027	98,550
2028	102,930
2029	107,675
2030	112,420
2031	117,530
2032	122,640

Organic waste type j fraction on wet basis:

Waste type j	Weight Fraction (wet waste)
W ₁ -Wood and wood products	0.0301
W ₂ -Pulp, paper and cardboard	0.0834
W ₃ -Food, food waste, beverages and tobacco	0.4027
W ₄ -Textiles	0.0304
W ₅ -Garden, yard and park waste	0.1341

	W ₆ -Glass, plastic, metal other inert	0.3193	
Justification of choice of data or description of measurement methods and procedures applied	It is calculated as total waste amount dumped in the landfill site in the year x multiplied by organic waste type j fraction on wet basis. Both total waste amount and waste type j fraction are referred to FSR.		
Purpose of Data	Calculation of baseline emissions		
Comments	-		

Data / Parameter	DOC _j													
Data unit	-													
Description	Fraction of degradable organic carbon in the waste type j (weight fraction)													
Source of data	“Emissions from solid waste disposal sites” (version 08.1)													
Value applied	<table><tr><td>Waste type j</td><td>DOC_j (% wet waste)</td></tr><tr><td>Wood and wood products</td><td>43</td></tr><tr><td>Pulp, paper and cardboard (other than sludge)</td><td>40</td></tr><tr><td>Textiles</td><td>24</td></tr><tr><td>Garden, yard and park waste</td><td>20</td></tr><tr><td>Glass, plastic, metal, other inert waste</td><td>0</td></tr></table>		Waste type j	DOC _j (% wet waste)	Wood and wood products	43	Pulp, paper and cardboard (other than sludge)	40	Textiles	24	Garden, yard and park waste	20	Glass, plastic, metal, other inert waste	0
Waste type j	DOC _j (% wet waste)													
Wood and wood products	43													
Pulp, paper and cardboard (other than sludge)	40													
Textiles	24													
Garden, yard and park waste	20													
Glass, plastic, metal, other inert waste	0													
Justification of choice of data or description of measurement methods and procedures applied	The climate condition ¹³ of location of Shangluo MSW treatment site: Temperature:7.8-13.9℃ Mean annual precipitation: 696.8-830.1mm Potential evapotranspiration: 1450.82mm													
Purpose of Data	Calculation of baseline emissions													
Comments	-													

¹³ The Feasibility Study Report of the project.

Data / Parameter	k_j															
Data unit	1/yr															
Description	Decay rate for the waste type j															
Source of data	“Emissions from solid waste disposal sites” (version 08.1)															
Value applied	<table border="1"> <tr> <td colspan="2">Waste type j</td><td>k_j ($MAT \leq 20^\circ C$, $MAP/PET < 1$)</td></tr> <tr> <td rowspan="2">Slowly degrading</td><td>Pulp, paper, cardboard (other than sludge), textiles</td><td>0.04</td></tr> <tr> <td>Wood, wood products and straw</td><td>0.02</td></tr> <tr> <td>Moderately degrading</td><td>Other(non-food) organic putrescible garden and park waste</td><td>0.05</td></tr> <tr> <td>Rapidly degrading</td><td>Food, food waste, sewage sludge, beverages and tobacco</td><td>0.06</td></tr> </table>		Waste type j		k_j ($MAT \leq 20^\circ C$, $MAP/PET < 1$)	Slowly degrading	Pulp, paper, cardboard (other than sludge), textiles	0.04	Wood, wood products and straw	0.02	Moderately degrading	Other(non-food) organic putrescible garden and park waste	0.05	Rapidly degrading	Food, food waste, sewage sludge, beverages and tobacco	0.06
Waste type j		k_j ($MAT \leq 20^\circ C$, $MAP/PET < 1$)														
Slowly degrading	Pulp, paper, cardboard (other than sludge), textiles	0.04														
	Wood, wood products and straw	0.02														
Moderately degrading	Other(non-food) organic putrescible garden and park waste	0.05														
Rapidly degrading	Food, food waste, sewage sludge, beverages and tobacco	0.06														
Justification of choice of data or description of measurement methods and procedures applied	The climate condition¹⁴ of location of Shangluo MSW treatment site: Temperature: 7.8-13.9°C Mean annual precipitation: 696.8-830.1mm Potential evapotranspiration: 1450.82mm															
Purpose of Data	Calculation of baseline emissions															
Comments	-															

Data / Parameter	D_{CH_4}
Data unit	t/m ³
Description	Density of methane at the temperature and pressure of the landfill gas in the year y

¹⁴ The Feasibility Study Report of the project.

Source of data	Default value as per the latest version of “Project emissions from flaring”
Value applied	0.000716 at normal conditions
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO ₂ /MWh
Description	Operation margin emission factor of NCPG
Source of data	2023 Bulletin on the Baseline Emission Factors of the China Grids
Value applied	0.9014
Justification of choice of data or description of measurement methods and procedures applied	The value is published by Ministry of Ecology and Environment of the People’s Republic of China, based on China Energy Statistic Yearbook (2020-2022), System of Statistical Survey of Energy Consumption in Public Institutions (National Government Offices Administration, approved by the National Bureau of Statistics, August 2019), China Electric Power Yearbook (2020-2022), Power Industry Statistical Data Collection (2019-2021) and IPCC data.
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{grid,BM,y}$
Data unit	tCO ₂ /MWh
Description	Build margin emission factor of NCPG
Source of data	2023 Bulletin on the Baseline Emission Factors of the China Grids
Value applied	0.3597

Justification of choice of data or description of measurement methods and procedures applied	The value is published by Ministry of Ecology and Environment of the People's Republic of China, based on China Energy Statistic Yearbook (2020-2022), System of Statistical Survey of Energy Consumption in Public Institutions (National Government Offices Administration, approved by the National Bureau of Statistics, August 2019), China Electric Power Yearbook (2020-2022), Power Industry Statistical Data Collection (2019-2021) and IPCC data.
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO ₂ /MWh
Description	Combine margin emission factor of NCPG
Source of data	Calculated based on "Tool to calculate the emission factor for an electricity system (version 07.0)"
Value applied	0.63055
Justification of choice of data or description of measurement methods and procedures applied	The value is calculated based on $EF_{grid,OM,y}$ and $EF_{grid,BM,y}$ as per "Tool to calculate the emission factor for an electricity system (version 07.0)"
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	NCV _{CH4}
Data unit	MJ/Nm ³
Description	Net calorific value of methane at reference conditions
Source of data	AMS-III.G Landfill methane recovery (version 10.0)
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	-

Value applied	35.9
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	EE _y
Data unit	%
Description	Energy Conversion Efficiency of the project equipment
Source of data	Default value of AMS.III.G (Version 10.0)
Description of measurement methods and procedures to be applied	As per paragraph 21 of AMS.III.G (Version 10.0), there are two options to decide the value of EE _y . Option 1: Specification provided by the equipment manufacturer. The equipment shall be designed to utilize biogas as fuel, and the efficiency specification is for biogas. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation. Option 2: Default efficiency of 40 percent. Default value 40% is applied in the project.
Frequency of monitoring/recording	Not applicable. Because default value of AMS.III.G (Version 10.0) is adopted in the project.
Value applied	40%(default value)
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	$F_{CH_4, BL, y}$																								
Data unit	tCH ₄ /yr																								
Description	Amount of methane in the LFG which is flared due to a requirement in year y																								
Source of data	Information of the host country's regulatory requirements relating to LFG, contractual requirements, or requirements to address safety and odour concerns.																								
Description of measurement methods and procedures to be applied	According to AMS III.G, the relevant procedures in ACM0001 may be followed to determine parameter $F_{CH_4, BL, y}$. Case 2 of section 5.4.1.3 of ACM0001 "Flaring or use of landfill gas" (version 19.0) is applied for the project.																								
Frequency of monitoring/recording	Annually																								
Value applied	The data below are calculated ex ante: <table border="1"> <thead> <tr> <th>Year</th><th>Amount of methane in the LFG which is flared due to a requirement (tCH₄)</th></tr> </thead> <tbody> <tr> <td>23-Mar-2023 to 31-Dec-2023</td><td>150</td></tr> <tr> <td>01-Jan-2024 to 31-Dec-2024</td><td>204</td></tr> <tr> <td>01-Jan-2025 to 31-Dec-2025</td><td>216</td></tr> <tr> <td>01-Jan-2026 to 31-Dec-2026</td><td>228</td></tr> <tr> <td>01-Jan-2027 to 31-Dec-2027</td><td>240</td></tr> <tr> <td>01-Jan-2028 to 31-Dec-2028</td><td>253</td></tr> <tr> <td>01-Jan-2029 to 31-Dec -2029</td><td>267</td></tr> <tr> <td>01-Jan-2030 to 31-Dec-2030</td><td>280</td></tr> <tr> <td>01-Jan-2031 to 31-Dec-2031</td><td>295</td></tr> <tr> <td>01-Jan-2032 to 31-Dec-2032</td><td>310</td></tr> <tr> <td>01-Jan-2033 to 22-Mar-2033</td><td>65</td></tr> </tbody> </table>	Year	Amount of methane in the LFG which is flared due to a requirement (tCH ₄)	23-Mar-2023 to 31-Dec-2023	150	01-Jan-2024 to 31-Dec-2024	204	01-Jan-2025 to 31-Dec-2025	216	01-Jan-2026 to 31-Dec-2026	228	01-Jan-2027 to 31-Dec-2027	240	01-Jan-2028 to 31-Dec-2028	253	01-Jan-2029 to 31-Dec -2029	267	01-Jan-2030 to 31-Dec-2030	280	01-Jan-2031 to 31-Dec-2031	295	01-Jan-2032 to 31-Dec-2032	310	01-Jan-2033 to 22-Mar-2033	65
Year	Amount of methane in the LFG which is flared due to a requirement (tCH ₄)																								
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01-Jan-2033 to 22-Mar-2033	65																								
Monitoring equipment	-																								
QA/QC procedures to be applied	Yearly updated in line with the relevant local laws and regulation if such laws and regulations exist.																								

Purpose of data	Calculation of baseline emissions
Calculation method	The data is calculated as methane captured and destroyed multiplied by fraction of LFG that is required to be flared due to a requirement.
Comments	Applicable to Case 2 of section 5.4.1.3 of ACM0001 “Flaring or use of landfill gas” (version 19.0)

Data / Parameter	$\rho_{reg,y}$
Data unit	Dimensionless
Description	Fraction of LFG that is required to be flared due to a requirement in year y
Source of data	Information of the host country’s regulatory requirements relating to LFG, contractual requirements, or requirements to address safety and odour concerns.
Description of measurement methods and procedures to be applied	According to the methodology ACM0001(version 19.0), if the requirement does not specify any amount or percentage of LFG that should be destroyed, but requires the installation of a system to capture and flare of the LFG, then a typical destruction rate of 20 percent is assumed.
Frequency of monitoring/recording	Annually
Value applied	20%
Monitoring equipment	-
QA/QC procedures to be applied	Yearly updated in line with the relevant local laws and regulation if such laws and regulations exist.
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	Applicable to Case 2 of section 5.4.1.3 of ACM0001 “Flaring or use of landfill gas” (version 19.0)

Data / Parameter	$EG_{PJ, facility,y}$
Data unit	MWh

Description	Quantity of net electricity generated supplied by the project plant/unit in year y																								
Source of data	Electricity meter E2																								
Description of measurement methods and procedures to be applied	Measured continuously by electricity meter(bi-directional) installed at the project site. All data will be monitored and archived electronically. Double check by receipt of electricity sales.																								
Frequency of monitoring/recording	The recording frequency will be hourly measured and record, and monthly aggregated.																								
Value applied	The data below are calculated ex ante <table> <tr> <th>Year</th><th>Quantity of net electricity generated supplied by the project plant (MWh)</th></tr> <tr> <td>23-Mar-2023 to 31-Dec-2023</td><td>3,153</td></tr> <tr> <td>01-Jan-2024 to 31-Dec-2024</td><td>4,294</td></tr> <tr> <td>01-Jan-2025 to 31-Dec-2025</td><td>4,540</td></tr> <tr> <td>01-Jan-2026 to 31-Dec-2026</td><td>4,792</td></tr> <tr> <td>01-Jan-2027 to 31-Dec-2027</td><td>5,051</td></tr> <tr> <td>01-Jan-2028 to 31-Dec-2028</td><td>5,320</td></tr> <tr> <td>01-Jan-2029 to 31-Dec -2029</td><td>5,600</td></tr> <tr> <td>01-Jan-2030 to 31-Dec-2030</td><td>5,890</td></tr> <tr> <td>01-Jan-2031 to 31-Dec-2031</td><td>6,192</td></tr> <tr> <td>01-Jan-2032 to 31-Dec-2032</td><td>6,505</td></tr> <tr> <td>01-Jan-2033 to 22-Mar-2033</td><td>1,358</td></tr> </table>	Year	Quantity of net electricity generated supplied by the project plant (MWh)	23-Mar-2023 to 31-Dec-2023	3,153	01-Jan-2024 to 31-Dec-2024	4,294	01-Jan-2025 to 31-Dec-2025	4,540	01-Jan-2026 to 31-Dec-2026	4,792	01-Jan-2027 to 31-Dec-2027	5,051	01-Jan-2028 to 31-Dec-2028	5,320	01-Jan-2029 to 31-Dec -2029	5,600	01-Jan-2030 to 31-Dec-2030	5,890	01-Jan-2031 to 31-Dec-2031	6,192	01-Jan-2032 to 31-Dec-2032	6,505	01-Jan-2033 to 22-Mar-2033	1,358
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01-Jan-2026 to 31-Dec-2026	4,792																								
01-Jan-2027 to 31-Dec-2027	5,051																								
01-Jan-2028 to 31-Dec-2028	5,320																								
01-Jan-2029 to 31-Dec -2029	5,600																								
01-Jan-2030 to 31-Dec-2030	5,890																								
01-Jan-2031 to 31-Dec-2031	6,192																								
01-Jan-2032 to 31-Dec-2032	6,505																								
01-Jan-2033 to 22-Mar-2033	1,358																								
Monitoring equipment	Electricity meter E2 shown in the monitoring system as below.																								
QA/QC procedures to be applied	The calibration should be done once a year by a qualified third party.																								
Purpose of data	Calculation of baseline emissions																								
Calculation method	This parameter should be either monitored using bi-directional energy meter or calculated as difference between (a) the quantity of electricity supplied by the project plant/unit to the grid; and (b) the quantity of electricity the project plant/unit from the grid.																								

	In case it is calculated then the following parameters shall be measured: (a) The quantity of electricity supplied by the project plant/unit to the grid; and (b) The quantity of electricity delivered to the project plant/unit from the grid
Comments	-

Data / Parameter	EG _y																						
Data unit	MWh																						
Description	Electricity generation in year y																						
Source of data	Electricity meter E1																						
Description of measurement methods and procedures to be applied	Measured continuously by electricity meter installed at the project site. All data will be monitored and archived electronically.																						
Frequency of monitoring/recording	The recording frequency will be hourly measured and record, and monthly aggregated.																						
Value applied	The data below are calculated ex ante <table border="1"> <thead> <tr> <th>Year</th><th>Electricity generated in year y (MWh)</th></tr> </thead> <tbody> <tr> <td>23-Mar-2023 to 31-Dec-2023</td><td>3,354</td></tr> <tr> <td>01-Jan-2024 to 31-Dec-2024</td><td>4,568</td></tr> <tr> <td>01-Jan-2025 to 31-Dec-2025</td><td>4,830</td></tr> <tr> <td>01-Jan-2026 to 31-Dec-2026</td><td>5,097</td></tr> <tr> <td>01-Jan-2027 to 31-Dec-2027</td><td>5,374</td></tr> <tr> <td>01-Jan-2028 to 31-Dec-2028</td><td>5,660</td></tr> <tr> <td>01-Jan-2029 to 31-Dec -2029</td><td>5,958</td></tr> <tr> <td>01-Jan-2030 to 31-Dec-2030</td><td>6,266</td></tr> <tr> <td>01-Jan-2031 to 31-Dec-2031</td><td>6,587</td></tr> <tr> <td>01-Jan-2032 to 31-Dec-2032</td><td>6,920</td></tr> </tbody> </table>	Year	Electricity generated in year y (MWh)	23-Mar-2023 to 31-Dec-2023	3,354	01-Jan-2024 to 31-Dec-2024	4,568	01-Jan-2025 to 31-Dec-2025	4,830	01-Jan-2026 to 31-Dec-2026	5,097	01-Jan-2027 to 31-Dec-2027	5,374	01-Jan-2028 to 31-Dec-2028	5,660	01-Jan-2029 to 31-Dec -2029	5,958	01-Jan-2030 to 31-Dec-2030	6,266	01-Jan-2031 to 31-Dec-2031	6,587	01-Jan-2032 to 31-Dec-2032	6,920
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	01-Jan-2033 to 22-Mar-2033	1,445
Monitoring equipment	Electricity meter E1 shown in the monitoring system as below.	
QA/QC procedures to be applied	The calibration should be done once a year by a qualified third party.	
Purpose of data	Calculation of baseline emissions	
Calculation method	-	
Comments	-	

5.3 Monitoring Plan

1. The requirement of monitoring plan

The project participants will monitor the emission reductions by methods, indicators, and frequency required by Monitoring Methodology AMS-III.G (Version 10.0) and AMS-I.D (Version 18.0) to ensure project ERs are measurable and real. The monitoring methodology is based on measurement of the amount of electricity generation and quantity of net electricity generation supplied by the project.

2. Responsibilities of operational and management structure

The project owner will implement this monitoring plan. The plan could be revised according to suggestions from VVB and the practical circumstances, in order to keep it consistent, transparent and conservative during the monitoring process.

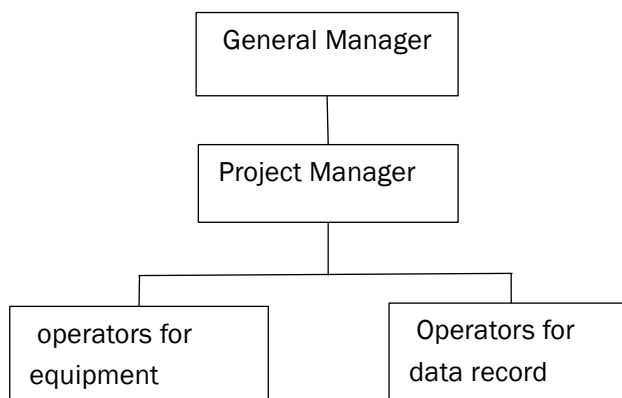


Figure 5.1 Operational and management structure

(1) Principal of the monitoring procedure

The general manager of the project is the leader of the monitoring tasks: he sets out the responsibility of everyone in the monitoring system, and establishes the related documents. The general manager ensures that staff in the monitoring system has the ability to deal with the assigned tasks.

(2) Executive person of the monitoring procedure

Project Manager: a Project Manager is appointed, specifically responsible for training, checking the daily operation, reporting forms and archiving emergency situation reports. The Project Manager reports monthly to the General Manager (GM) about the project performance and monitored data. In the event that non-conformance in the performance to the mentioned procedures and/or functioning problems of the monitoring equipment are identified, the Project manager will inform the GM about the situation and work out relevant measures to be taken. The Project manager will also be responsible for aggregating the monitored data monthly and yearly, archiving and keeping data during the crediting period and two years after.

(3) Operators of the monitoring procedure

Operators will take turns to work in the control center 24 hours a day. They will be in charge of data supervision, filling operation report forms and checking and inspecting the system. If necessary, they will have the responsibility for executing the emergency plan and drafting emergency situation reports.

3. Monitoring system¹⁵

¹⁵ During normal operation, the project consumes a small amount of its own generated renewable electricity for plant production. The remaining electricity will be delivered to the grid.

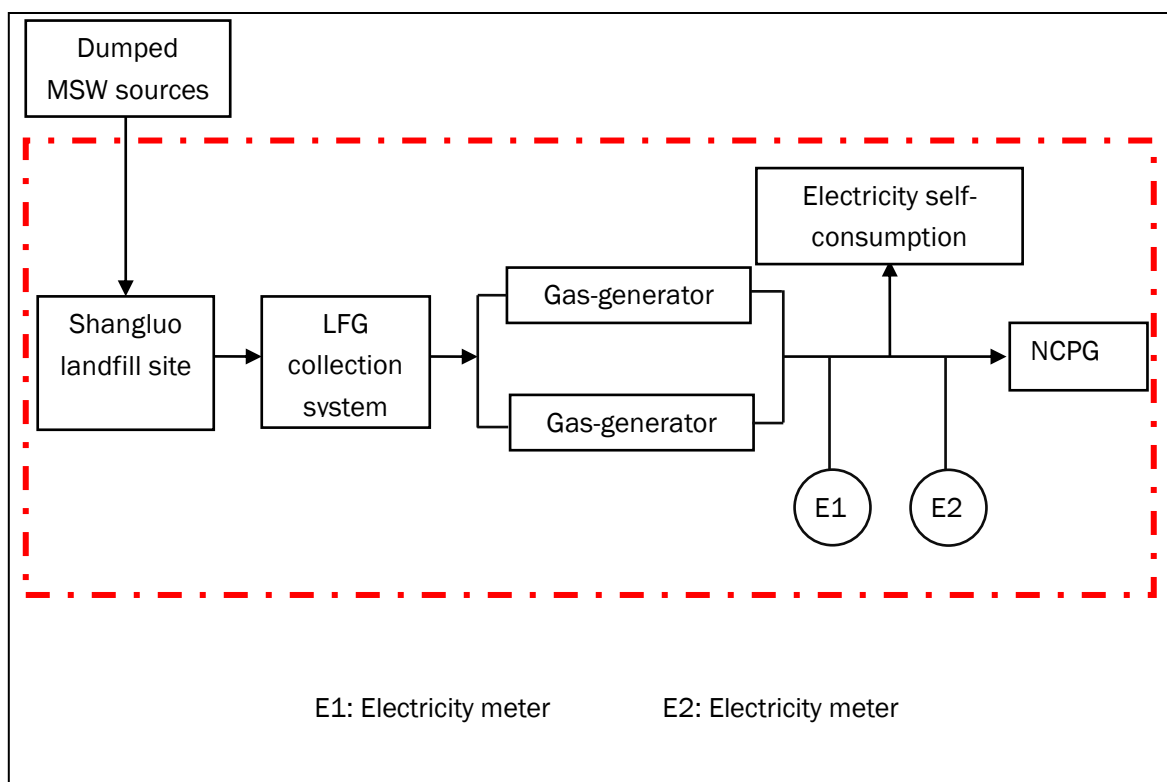


Figure 5.2 Monitoring system

Main parameters:

E1	Electricity meter, continuously measure the total electricity by all the generators
E2	Electricity meter (bi-directional), continuously measure the electricity supplied to the NCPG and the electricity supplied by the grid

4. Data collection and management

The monitoring frequency is continuous. The data will be recorded and electronic archived by computer automatically. The operators will check the data everyday, and totalize the data every month. The records will be submitted to the GM manager monthly, and the operator in charge of data record will keep a set of copy ones. All these records shall be verified by the GM manager and kept up to 2 years after the end of the crediting period.

5. Calibration

In order to ensure the reliability of the data measured, the monitoring instruments will be calibrated as per state and/or sector standards and rules and usually calibrated each year.

The operator who is responsible for equipment maintenance invites a qualified third party to calibrate the monitoring meters listed above. The calibration reports are checked by the project manager and then archived in the project owner archives by the operator.

6. QA/QC

All the monitoring devices listed above will be calibrated once a year by a qualified third party. The calibration report will be kept in the archives by the project owner, and checked by Project Manager. When the data is not available in case of any emergency cases such as failure of meters, the emission reductions in this period will not be claimed. It will be checked immediately, necessary standby meters will be installed when the monitoring meters are not functioning.

The project performance and monitoring data are reported monthly to the General Manager for review and internal audit of calculation and archived and kept for emission reduction calculation yearly during the crediting period and two years after.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

No commercially sensitive information included in the project description to be excluded in the public version.

Section	Information	Justification
/	/	/